

# Appendix

This appendix is the formal specification of FORGE’s cost–benefit methodology that backs the Methods and case studies in the main text. Each section below lists variables, units, equations, and conditional logic for one cost or benefit component so the mapping from engineering assumptions to dollar totals is transparent. All cost and benefit symbols ( $C$ ,  $B$ ) denote present values unless explicitly marked ,nominal.

**Conventions.** Only hard capital costs—build (§B.1.1), ROW acquisition (§B.1.2), and environmental mitigation (§B.1.3)—are AFUDC-eligible and enter rate base at COD (§A.3). All other costs (O&M, insurance, ROW rent, energy losses, delay, risk, emissions) and all benefits are neither AFUDC-eligible nor part of rate base.

**Project type.** Every FORGE scenario specifies a **project type** that determines how several cost and benefit modules branch. Three values are defined:

| Project type   | Description                                                                                                                                                                                                                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Greenfield     | New transmission line on a new or existing corridor. Full structure and converter costs; full environmental mitigation credits; capacity relief uses flow-factor formula ( $\phi \times C_{\text{new}}$ ); standard voltage class for energy losses.                                                                                |
| Reconductoring | Conductor replacement on existing structures. Structure and converter costs zeroed; environmental mitigation credits zeroed (no ground disturbance); capacity relief uses incremental formula ( $C_{\text{new}} - C_{\text{old}}$ ); energy losses use old-voltage/new-resistance hybrid lookup (towers unchanged).                 |
| Rebuild        | Full structure and conductor replacement on an existing corridor. Full structure and converter costs; environmental mitigation credits scaled by $f_{\text{rebuild,env}}$ (default 0.5); capacity relief uses incremental formula ( $C_{\text{new}} - C_{\text{old}}$ ); standard voltage class for energy losses (new structures). |

Both reconductoring and rebuild require  $C_{\text{old}}$  (old capacity). For greenfield,  $C_{\text{old}}$  is not used.

## A Utilities

### A.1 Weighted Miles

Default terrain multipliers  $\lambda_{\text{terrain}}$  are sourced from the WECC Transmission Expansion Planning cost study (Rawlins et al. 2014), which derives construction difficulty factors from utility-reported project data (Black & Veatch, 2014). The wetland multiplier (2.0) is updated from MISO MTEP25 matting and construction-difficulty adders (MISO 2025b), superseding the earlier WECC value (1.20).

#### Variables

| Variable                      | Description                                                                                                                                                       |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $M_{\text{terrain}}$          | Miles of a specific terrain type (miles). Terrain set as defined in this subsection. Only terrains with $M_{\text{terrain}} > 0$ are used in sums.                |
| $\lambda_{\text{terrain}}$    | Cost multiplier by terrain type (dimensionless, $\geq 1$ ). Same terrain list as $M_{\text{terrain}}$ .                                                           |
| $M_{\text{total}}$            | Total (raw) miles (miles). Sum of $M_{\text{terrain}}$ over all terrains.                                                                                         |
| $M_{\text{weighted,terrain}}$ | Weighted miles for that terrain type (miles). $M_{\text{weighted,terrain}} = M_{\text{terrain}} \times \lambda_{\text{terrain}}$ for that terrain.                |
| $M_{\text{weighted,total}}$   | Total weighted miles across all terrain types (miles). Used in Build costs (§B.1.1) for conductor/structure/converter variable costs.                             |
| $\lambda_{\text{average}}$    | Average terrain multiplier for the project (dimensionless). $\lambda_{\text{average}} = M_{\text{weighted,total}}/M_{\text{total}}$ when $M_{\text{total}} > 0$ . |

#### Equations

##### 1. Weighted miles for each terrain type

$$M_{\text{weighted,terrain}} = M_{\text{terrain}} \times \lambda_{\text{terrain}} \quad \text{for that terrain type.}$$

##### 2. Total weighted miles

$$M_{\text{weighted,total}} = \sum_{\text{terrain}} M_{\text{weighted,terrain}}.$$

##### 3. Total (raw) miles

$$M_{\text{total}} = \sum_{\text{terrain}} M_{\text{terrain}}.$$

#### 4. Average terrain multiplier

$$\lambda_{\text{average}} = \frac{M_{\text{weighted,total}}}{M_{\text{total}}} \quad (M_{\text{total}} > 0).$$

Table 1: Default terrain multipliers  $\lambda_{\text{terrain}}$ .

| Terrain       | $\lambda$ | Source                |
|---------------|-----------|-----------------------|
| Forested      | 2.25      | WECC 2014             |
| Scrubbed flat | 1.00      | WECC 2014 (baseline)  |
| Wetland       | 2.00      | MISO MTEP25           |
| Farmland      | 1.00      | WECC 2014 (baseline)  |
| Desert/barren | 1.05      | WECC 2014             |
| Urban         | 1.59      | WECC 2014             |
| Rolling hills | 1.40      | WECC 2014             |
| Mountain      | 1.75      | WECC 2014             |
| Subsea        | 1.00      | Marine costs separate |

## A.2 Financial Parameters

The default social discount rate of 2% aligns with Office of Management and Budget (2023), which derives its recommended rate empirically from long-term Treasury yields, and with the Ramsey discounting framework in Rennert et al. (2022).

### Variables

| Variable                  | Description                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\pi$                     | Annual inflation rate (decimal). Used in Fisher link for real WACC. Default 3% (Social Security Board of Trustees 2025).                                                                                                                                                                                                                                              |
| $Y_{\text{base}}$         | Base year (year). Reference year for all present values and cost parameters. Default 2025; all costs expressed in $Y_{\text{base}}$ constant dollars.                                                                                                                                                                                                                 |
| $r_{\text{WACC,nom}}$     | Nominal weighted average cost of capital (decimal). Direct input. Converted to $r_{\text{WACC,real}}$ via Fisher link; not used directly in any calculation. Default 7.5% (California Public Utilities Commission 2025).                                                                                                                                              |
| $r_{\text{WACC,real}}$    | Real weighted average cost of capital (decimal). Used for societal PV of costs and benefits. Calculated via Fisher link.                                                                                                                                                                                                                                              |
| $r_{\text{social}}$       | Social discount rate (decimal). Used for externality costs (emissions, expected wildfire cost, expected outage cost).                                                                                                                                                                                                                                                 |
| $g_{\text{construction}}$ | Construction cost escalation rate (decimal). Default 4%/yr nominal. Applied during delay (§B.1.1). Source: MISO MTEP25 (MISO 2025a).                                                                                                                                                                                                                                  |
| $g_{\text{benefit}}$      | Benefit price escalation rate (decimal). Real annual escalation applied to congestion and delivered-energy benefit prices. Default 1%/yr real. At 0%, reduces to the flat-price case. Source: Chang, Pfeifenberger, and Hagerty (2013) (1% real tail-extension practice); California Independent System Operator (2025) (0% real current practice); sensitivity 0–4%. |
| $g_{\text{rent}}$         | ROW rent escalation rate (decimal). Real annual escalation applied to operational ROW rent (§B.2.3). Default 0%/yr real. Bureau of Land Management (BLM) adjusts at the Implicit Price Deflator for Gross Domestic Product (IPD–GDP) rate ( $\approx$ inflation); no above-inflation trend (Bureau of Land Management 2026a; U.S. Government 2024).                   |

### Equations

#### 1. Real WACC (Fisher link)

$$r_{\text{WACC,real}} = \frac{1 + r_{\text{WACC,nom}}}{1 + \pi} - 1.$$

#### When each rate is used

- $r_{\text{WACC,nom}}$ : Specified as a model input; not used directly in any calculation. All calculations—including AFUDC—use  $r_{\text{WACC,real}}$  derived via the Fisher link above.
- $r_{\text{WACC,real}}$ : Societal PV of capital, O&M, insurance, ROW rent, energy loss costs, benefits, revenue; delay and congestion delay costs.
- $r_{\text{social}}$ : Societal PV of externality costs: emissions (§B.4), expected cost of wildfires (§B.3.1), expected cost of outages (§B.3.2).

- $g_{\text{benefit}}$ : Real annual growth in benefit prices (congestion, delivered energy) and welfare-consistent energy valuation (thermal loss cost, delay costs). Applied from COD over  $T_{\text{lifetime}}$  in §C.1.1, §C.2.1, §B.2.5; applied over  $T_{\text{delay}} + T_{\text{construction}}$  in §B.2.7.
- $g_{\text{rent}}$ : Real annual growth in ROW rent. Applied from  $t_{\text{rent,start}}$  over  $T_{\text{rent}}$  in §B.2.3.

**Cost normalization.** All cost parameters in this appendix are expressed in  $Y_{\text{base}}$  constant dollars (default 2025). Source values from earlier or later years are escalated or deflated using the Bureau of Labor Statistics (BLS) Consumer Price Index for All Urban Consumers (CPI-U, Series CUUR0000SA0, annual averages, 1982–84 = 100). Table 2 lists the adjustment factors applied. Items accepted on author judgment (wildfire severity composite, urban vegetation management) are flagged in their respective subsections.

Table 2: Base-year normalization factors (CPI-U annual averages).

| Source year | CPI-U          | Factor → 2025\$ | Parameters                                                           |
|-------------|----------------|-----------------|----------------------------------------------------------------------|
| 2010        | 218.056        | 1.4762          | Habitat conservation credits                                         |
| 2008        | 215.303        | 1.4951          | Vegetation management (desert, farmland)                             |
| 2020        | 258.811        | 1.2438          | Social cost of CO <sub>2</sub> , SO <sub>x</sub> , NO <sub>x</sub>   |
| 2021        | 270.970        | 1.1880          | Outage redispatch cost                                               |
| 2022        | 292.655        | 1.0999          | Vegetation management (rolling hills)                                |
| 2023        | 304.702        | 1.0564          | Non-overhead build costs (IET)                                       |
| ~2023.5     | 309.194        | 1.0411          | Wildfire severity (composite)                                        |
| 2024        | 313.685        | 1.0262          | Value of lost load tiers, vegetation management (forested, scrubbed) |
| <b>2025</b> | <b>321.900</b> | <b>1.0000</b>   | Target base year                                                     |
| 2026        | 330.900        | 0.9728          | BLM ROW rents and hold costs (deflation)                             |

Overhead conductor, structure, and converter-station costs are sourced directly from MISO MTEP25 (MISO 2025b) at 2025\$ and do not require index adjustment. BLM acquisition costs (15 zones) are exempt from normalization: they are administrative zone values set by BLM regulation, unchanged since 2016, and represent actual schedule amounts payable (Bureau of Land Management 2026b). Environmental mitigation base costs for forested (\$5,486/acre) and mountainous (\$7,132/acre) terrain use MISO MTEP25 clearing costs directly at 2025\$ (MISO 2025b). Remaining terrain base mitigation costs (scrubbed flat, wetland, farmland, desert/barren, urban, rolling hills: \$800–\$5,000/acre overhead; underground values scaled 1–2× for trench/restoration scope) are order-of-magnitude estimates anchored to MISO Cost Estimation Guide clearing benchmarks (MISO 2025a). Published sources lack a terrain×construction-type matrix for transmission environmental restoration; per-terrain differentials reflect author judgment informed by relative clearing difficulty. Underground wetland (\$10,000/acre, author estimate) reflects dewatering and wetland-restoration requirements beyond surface clearing. Habitat conservation credits default to \$110,024/acre (2025\$), sourced from the Ecosystem Marketplace national average (Madsen, Carroll, and Moore Brands 2010). The uniform rate across terrains reflects that credit prices vary primarily by geography (U.S. Government Accountability Office 2024).

The construction cost escalation rate  $g_{\text{construction}}$  is applied during the delay period (§B.1.1) to reflect that deferred projects face higher material, labor, and equipment costs when they eventually break ground.

### A.3 AFUDC and Rate Base

AFUDC compounds base-year spending on construction work in progress (CWIP) to the commercial operation date (COD). Rate base at COD is the sum of capitalized AFUDC-eligible capital: build, ROW capital (acquisition), and environmental mitigation. The real WACC from §A.2 is used as the AFUDC rate, because all FORGE costs are expressed in base-year (real) dollars. This approximates the traditional regulatory approach, which separately compounds debt and equity components at their respective rates. Using a blended WACC slightly over- or under-states the AFUDC balance depending on capital structure, but the difference is small.

## Variables

| Variable                  | Description                                                                                                                                               |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| $i$                       | Component index (index). $i \in \{\text{build, ROW, env mitigation}\}$ .                                                                                  |
| $C_i^{\text{nom}}$        | Spending (pre-AFUDC) for component $i$ , base-year (\$). In base-year real dollars.                                                                       |
| $T_{\text{delay}}$        | Delay years (years). From project technical details.                                                                                                      |
| $T_{\text{construction}}$ | Construction years (years). From project technical details.                                                                                               |
| $w_y$                     | Spending weight for construction year $y$ (dimensionless). From S-curve profile; $\sum_{y=1}^{T_{\text{construction}}} w_y = 1$ .                         |
| $r_{\text{AFUDC}}$        | AFUDC rate (annual) (decimal). From §A.2: $r_{\text{AFUDC}} = r_{\text{WACC,real}}$ .                                                                     |
| $\text{AFUDC}_i$          | AFUDC allowance for component $i$ (\$). $C_i^{\text{cap}} - C_i^{\text{nom}}$ .                                                                           |
| $C_i^{\text{cap}}$        | Capitalized cost at COD for component $i$ (\$). $C_i^{\text{cap}} = C_i^{\text{nom}} + \text{AFUDC}_i$ .                                                  |
| $RB_{\text{real}}$        | Rate base at COD, base-year (\$). Sum of $C_i^{\text{cap}}$ over build, ROW, env mitigation. Already in base-year dollars (AFUDC compounds at real rate). |

## Equations

### 1. AFUDC rate.

$$r_{\text{AFUDC}} = r_{\text{WACC,real}} \quad (\text{from §A.2}).$$

**2. Capitalization to COD.** All AFUDC-eligible capital costs are incurred during the construction period. Each year’s fractional spend is compounded from its midpoint to COD using S-curve weights. See paragraph 2b.

**Delay-period spending.** ROW holding costs (annual option fees to maintain land access during the permitting period) are incurred during the delay period but are not AFUDC-eligible: they are operating expenses during delay and do not enter rate base. Only ROW acquisition is AFUDC-eligible capital. The cost timing pattern governs the delay/construction split for AFUDC-eligible costs; the default is 100% during construction.

**2b. Timing methodology (year-by-year compounding).** Construction spending follows an S-curve disbursement profile derived from MISO’s Transmission Cost Estimation Guide (MISO 2025a). Table 3 gives the reference profiles for 5–8 year construction durations; for other durations, the profiles are pooled into a normalized cumulative curve and resampled via piecewise linear interpolation. Each year’s fractional spend is compounded from its midpoint to COD:

$$C_i^{\text{cap}} = C_i^{\text{nom}} \sum_{y=1}^{T_{\text{construction}}} w_y (1 + r)^{T_{\text{construction}} - y + 0.5},$$

where  $w_y$  is the spending weight for year  $y$  ( $\sum w_y = 1$ ) and  $r = r_{\text{AFUDC}}$ . Hence  $\text{AFUDC}_i = C_i^{\text{cap}} - C_i^{\text{nom}}$ .

Table 3: MISO reference construction spending profiles (MISO 2025a).

| Duration | Yr 1 | Yr 2 | Yr 3 | Yr 4 | Yr 5 | Yr 6 | Yr 7 | Yr 8 |
|----------|------|------|------|------|------|------|------|------|
| 5-year   | 1%   | 4%   | 35%  | 35%  | 25%  |      |      |      |
| 6-year   | 1%   | 4%   | 10%  | 30%  | 35%  | 20%  |      |      |
| 7-year   | 1%   | 1%   | 3%   | 15%  | 30%  | 30%  | 20%  |      |
| 8-year   | 1%   | 1%   | 3%   | 5%   | 15%  | 25%  | 30%  | 20%  |

### 3. Rate base at COD.

$$RB_{\text{real}} = C_{\text{build}}^{\text{cap}} + C_{\text{ROW}}^{\text{cap}} + C_{\text{env mit}}^{\text{cap}}.$$

**Which costs enter rate base.** Build costs (§B.1.1), capital ROW (§B.1.2), and environmental mitigation (§B.1.3) are the three components in rate base. Capital recovery (§E) uses  $RB_{\text{real}}$  to compute the depreciation-plus-return component of the utility’s revenue requirement.

## A.4 System Constraints and Relief

### Variables

| Variable                       | Description                                                                                    |
|--------------------------------|------------------------------------------------------------------------------------------------|
| $\Delta C_{\text{effective}}$  | Effective capacity relief (MW).                                                                |
| $\phi$                         | Flow factor (dimensionless [0,1]). Greenfield: fraction of nameplate that relieves constraint. |
| $C_{\text{new}}$               | New line capacity (MW). All project types.                                                     |
| $C_{\text{old}}$               | Old line capacity (MW). Reconductoring and rebuild only (\$).                                  |
| $H$                            | Constrained hours (h/year). Hours/year the transmission constraint binds.                      |
| $X$                            | Average exceedance (MW). Average MW exceedance during constrained hours.                       |
| $\gamma_{\text{congestion}}$   | Value of congestion (redispatch cost) (\$/MWh).                                                |
| $B_{\text{congestion,annual}}$ | Annual congestion benefit (\$/year).                                                           |

### Equations

#### 1. Effective relief.

$$\Delta C_{\text{effective}} = \begin{cases} \phi \times C_{\text{new}} & \text{if greenfield} \\ C_{\text{new}} - C_{\text{old}} & \text{if reconductoring or rebuild} \end{cases}$$

**2. Single-constraint congestion relief.** During constrained hours  $H$ , the line relieves redispatch costs up to the lesser of effective capacity relief and average exceedance:

$$B_{\text{congestion,annual}} = H \times \min(\Delta C_{\text{effective}}, X) \times \gamma_{\text{congestion}}.$$

Remedial benefit equals congestion relief only. Curtailment is not modeled as a separate benefit category; the energy value of MWh that would have been curtailed is captured by delivered energy (§C.2.1).

## B Costs

This section defines the four cost categories: hard costs ( $C_{\text{hard}}$ ), soft costs ( $C_{\text{soft}}$ ), risk costs ( $C_{\text{risk}}$ ), and emissions costs ( $C_{\text{emissions}}$ ). The grand total cost for scenario  $P$  is  $C^P = C_{\text{hard}}^P + C_{\text{soft}}^P + C_{\text{risk}}^P + C_{\text{emissions}}^P$ .

### B.1 Hard Costs ( $C_{\text{hard}}$ )

Hard costs are one-time capital expenditures that are AFUDC-eligible and enter rate base at COD: build costs, capital ROW (acquisition), and environmental mitigation (see §A.3). These three components produce the hard cost category  $C_{\text{hard}}$  (§B.1.3).

#### B.1.1 Build costs

Default contingency is 20%, consistent with MISO’s planning-level estimate classification (MISO 2025a). A soft cost multiplier (default 10%) is applied after contingency to capture project management, engineering/design, and administrative costs. Component costs use weighted miles (§A.1); reconductoring projects exclude structure and converter costs (existing structures retained).

#### Default calibration

Overhead conductor variable costs (\$/mile) are derived directly from MISO MTEP25 (MISO 2025b) at native 2025\$: material, installation, and accessories from Tables 2.2-11/2.2-12 (Standard Aluminum ACSR/ACSS) and Table 2.2-13 (Advanced Aluminum/ACCC), plus OPGW and shieldwire from Table 2.2-14, using conductor selection and bundle counts from Tables 3.1-5 (AC) and 3.1-6 (HVDC). Per-1,000-ft unit costs convert to per-mile by  $\times 5.28$  (5,280/1,000); phase conductor costs scale by  $n_{\text{wires}} = (\text{conductors per phase}) \times (\text{phases}) \times (\text{circuits/poles})$ ; OPGW/shieldwire use MTEP25’s two-OPGW + one-shieldwire package once per mile. Standard Aluminum follows Table 3.1-5 conductor type (ACSS for 69–345 kV; ACSR for 500–765 kV and HVDC). ACCC sizes present in Table 2.2-13 are used directly; sizes absent from that table (Matt Skiles’ 430/611/725 kmil pairings) are piece-wise linearly interpolated or extrapolated on the MTEP25 ACCC cost curve by kmil. Overhead structure costs use MISO MTEP25 (MISO 2025b), directly in 2025\$. Converter-station fixed costs are one-end component sums from Tables 2.4-1 (LCC) and 2.4-2 (VSC) at native 2025\$; FORGE multiplies by  $N_{\text{conv}}$  (typically 2 for DC) at application time. Non-overhead (underground and subsea) conductor and fixed costs derive from the IET/Mott MacDonald Electricity Transmission Costing Study (Institution of Engineering and Technology and Mott MacDonald 2025), converted to 2025\$.

| Variable                           | Description                                                                                                                       |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| $C_{\text{conductor}}$             | Conductor cost (\$). Not terrain adjusted, no contingency applied.                                                                |
| $C_{\text{structure}}$             | Structure cost (\$). Not terrain adjusted, no contingency applied.                                                                |
| $C_{\text{converter}}$             | Converter cost (if DC) (\$). Not terrain adjusted, no contingency applied.                                                        |
| $C_{\text{adj,conductor}}$         | Conductor cost (\$). Terrain adjusted, no contingency applied.                                                                    |
| $C_{\text{adj,structure}}$         | Structure cost (\$). Terrain adjusted, no contingency applied.                                                                    |
| $C_{\text{adj,converter}}$         | Converter cost (if DC) (\$). Terrain adjusted, no contingency applied.                                                            |
| $C_{\text{adj,cont,conductor}}$    | Conductor cost (\$). Terrain adjusted, contingency applied.                                                                       |
| $C_{\text{adj,cont,structure}}$    | Structure cost (\$). Terrain adjusted, contingency applied.                                                                       |
| $C_{\text{adj,cont,converter}}$    | Converter cost (if DC) (\$). Terrain adjusted, contingency applied.                                                               |
| $F_{\text{conductor}}$             | Fixed conductor costs (\$). Conductors have both fixed and variable costs.                                                        |
| $F_{\text{structure}}$             | Fixed structure costs (\$). Currently, structures have no fixed costs in FORGE.                                                   |
| $F_{\text{converter}}$             | Fixed converter costs (\$). Converters only have fixed costs.                                                                     |
| $N_{\text{conv}}$                  | Number of converter stations (count). 0 for AC; typically 2 for DC. Applied at converter cost.                                    |
| $V_{\text{conductor}}$             | Variable conductor costs (\$/mile). Miles used are terrain adjusted (§A.1).                                                       |
| $V_{\text{structure}}$             | Variable structure costs (\$/mile). Miles used are terrain adjusted.                                                              |
| $V_{\text{converter}}$             | Variable converter costs (\$/mile). Converters currently have no variable costs — only fixed.                                     |
| $M_{\text{weighted,total}}$        | Total weighted miles (miles). From §A.1.                                                                                          |
| $r_{\text{contingency,conductor}}$ | Contingency multiplier conductors (dimensionless [0,1]).                                                                          |
| $r_{\text{contingency,structure}}$ | Contingency multiplier structures (dimensionless [0,1]).                                                                          |
| $r_{\text{contingency,converter}}$ | Contingency multiplier converter (dimensionless [0,1], if DC).                                                                    |
| $s_{\text{soft}}$                  | Soft cost multiplier (decimal fraction). Default 0.10 (10%). Applied after contingency. Covers PM, engineering, A&G (MISO 2025a). |
| $g_{\text{construction}}$          | Construction cost escalation rate (decimal). Default 4%/yr nominal (§A.2). Applied during delay.                                  |
| $C_{\text{build}}$                 | Total build cost, nominal (\$). Terrain adjusted, contingency applied.                                                            |
| $C_{\text{build}}$                 | Total build cost (PV) (\$). Societal perspective; spending spread over construction and discounted at $r_{\text{WACC,real}}$ .    |
| $T_{\text{delay}}$                 | Delay years (years). From project technical details.                                                                              |
| $T_{\text{construction}}$          | Construction years (years). From project technical details.                                                                       |
| $t_{\text{start}}$                 | Construction start year (year index). $t_{\text{start}} = T_{\text{delay}} + 1$ (1-based).                                        |
| $r_{\text{WACC,real}}$             | Real WACC (decimal). Societal discount rate (§A.2).                                                                               |
| $\xi_{\text{DC}}$                  | Binary for DC projects (0 or 1). 1 if DC, 0 if AC.                                                                                |
| $\xi_{\text{structure}}$           | Binary: structure cost applies (0 or 1). 0 if reconductoring (existing structures retained); 1 if greenfield or rebuild.          |

## Variables

## Equations

### 1. Component costs (raw miles).

$$C_{\text{conductor}} = V_{\text{conductor}} \cdot M_{\text{total}} + F_{\text{conductor}},$$

$$C_{\text{structure}} = V_{\text{structure}} \cdot M_{\text{total}} + F_{\text{structure}},$$

$$C_{\text{converter}} = V_{\text{converter}} \cdot M_{\text{total}} + F_{\text{converter}}.$$

In FORGE,  $F_{\text{structure}} = 0$  and  $V_{\text{converter}} = 0$ .

### 2. Terrain-adjusted component costs.

$$C_{\text{adj,conductor}} = V_{\text{conductor}} \cdot M_{\text{weighted,total}} + F_{\text{conductor}},$$

$$C_{\text{adj,structure}} = V_{\text{structure}} \cdot M_{\text{weighted,total}},$$

$$C_{\text{adj,converter}} = N_{\text{conv}} \cdot F_{\text{converter}}.$$

### 3. Contingency-applied costs.

$$C_{\text{adj,cont,conductor}} = (1 + r_{\text{contingency,conductor}}) \cdot C_{\text{adj,conductor}},$$

$$C_{\text{adj,cont,structure}} = (1 + r_{\text{contingency,structure}}) \cdot C_{\text{adj,structure}},$$

$$C_{\text{adj,cont,converter}} = (1 + r_{\text{contingency,converter}}) \cdot C_{\text{adj,converter}}.$$

**3b. Soft-cost-adjusted costs.** A soft cost multiplier captures project management, engineering/design, and administrative and general costs not otherwise itemized:

$$C_{\text{adj,cont,conductor}} \times = (1 + s_{\text{soft}}), \quad C_{\text{adj,cont,structure}} \times = (1 + s_{\text{soft}}), \quad C_{\text{adj,cont,converter}} \times = (1 + s_{\text{soft}}).$$

The default  $s_{\text{soft}} = 0.10$  (10%) is consistent with MISO MTEP planning methodology (MISO 2025a) and AACE 96R-18 owner's-cost/project-indirect-cost categories.

**3c. Delay escalation (if  $T_{\text{delay}} > 0$ ).** Construction costs escalate during the delay period at the construction cost escalation rate (adopted from MISO’s year-over-year general escalation of 4%/yr nominal (MISO 2025a)), reflecting material, labor, and equipment cost growth between project conception and construction start:

$$C_{\text{adj,cont,conductor}} \times = (1 + g_{\text{construction}})^{T_{\text{delay}}},$$

$$C_{\text{adj,cont,structure}} \times = (1 + g_{\text{construction}})^{T_{\text{delay}}},$$

$$C_{\text{adj,cont,converter}} \times = (1 + g_{\text{construction}})^{T_{\text{delay}}}.$$

At the default  $g_{\text{construction}} = 2.5\%/yr$  and a 5-year delay, construction costs increase by  $(1.025)^5 = 1.13\times$  (13%). For SunZia’s 17-year delay:  $(1.025)^{17} = 1.52\times$  (52% cost increase).

#### 4. Total build cost.

$$C_{\text{build}} = C_{\text{adj,cont,conductor}} + \xi_{\text{structure}} \cdot C_{\text{adj,cont,structure}} + \xi_{\text{structure}} \cdot \xi_{\text{DC}} \cdot C_{\text{adj,cont,converter}}.$$

**Regulatory perspective (AFUDC).** Build cost is AFUDC-eligible and is capitalized to COD using year-by-year S-curve compounding (see §A.3).

**Societal perspective (present value).** Build spending is spread evenly over  $T_{\text{construction}}$  starting at  $t_{\text{start}} = T_{\text{delay}} + 1$  and discounted at real WACC. For  $T_{\text{construction}} > 0$ :

$$C_{\text{build,annual}} = \frac{C_{\text{build}}}{T_{\text{construction}}}, \quad C_{\text{build}} = \sum_{t=t_{\text{start}}}^{t_{\text{start}}+T_{\text{construction}}-1} \frac{C_{\text{build,annual}}}{(1 + r_{\text{WACC,real}})^t}.$$

If  $T_{\text{construction}} = 0$ , the full cost is one-time at  $t_{\text{start}}$ :  $C_{\text{build}} = C_{\text{build}}/(1 + r_{\text{WACC,real}})^{t_{\text{start}}}$ .

#### B.1.2 Capital ROW costs (acquisition, holding)

Only acquisition is AFUDC-eligible and enters rate base at COD; holding is an operating expense during the delay period and does not enter rate base. ROW rent is an operational—not capital—cost (§B.2.3). ROW here refers only to the linear corridor (miles  $\times$  width by zone). FORGE omits converter station acreage.

##### Default calibration

Default acquisition and rent values ( $p_{\text{acquisition,zone}}$ ,  $p_{\text{rent,zone}}$ ) reflect the BLM Per-Acre Linear Rent Schedule for federal lands (Bureau of Land Management 2026b). These defaults apply to federal land; for private land, substitute state-level or appraised private-land values (e.g. USDA NASS county-level farmland values, state PUC filings, or project-specific appraisals). Per-zone acquisition and rent parameters may be varied in sensitivity analysis when project-specific land values differ from the BLM schedule.

| Variable                         | Description                                                                                                                     |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| zone                             | Zone index (index). ROW split into zones (1–15). Use only zones where $M_{\text{zone}} > 0$ .                                   |
| $M_{\text{zone}}$                | Miles in zone (miles).                                                                                                          |
| $W_{\text{ROW}}$                 | ROW width (ft). Contingent on project category.                                                                                 |
| $A_{\text{zone}}$                | Area of ROW of zone (acres).                                                                                                    |
| $p_{\text{acquisition,zone}}$    | Acquisition cost per acre, zone (\$/acre). One-time.                                                                            |
| $p_{\text{hold,zone}}$           | Holding (option fee) per acre per year, zone (\$/acre/year).                                                                    |
| $p_{\text{rent,zone}}$           | Rent per acre per year, zone (\$/acre/year). Operational only (§B.2.3).                                                         |
| $C_{\text{acquisition}}$         | Total one-time acquisition cost (\$). From zone sum.                                                                            |
| $C_{\text{hold,annual}}$         | Total annual holding cost (\$/year). Permitting/delay years.                                                                    |
| $T_{\text{delay}}$               | Delay years (years). Used for holding.                                                                                          |
| $C_{\text{hold,total}}$          | Total holding cost over delay (\$). Option fee over full delay period.                                                          |
| $\xi_{\text{acquisition}}$       | Binary: acquisition counts toward capital ROW (0 or 1). 1 if acquisition applies, 0 if not (e.g. lease). Set by agreement type. |
| $\xi_{\text{holding}}$           | Binary: holding counts toward capital ROW (0 or 1). 1 if holding applies, 0 if not. Set by agreement type.                      |
| $C_{\text{acquisition}}$         | PV of acquisition (\$). Societal; discounted at $r_{\text{WACC,real}}$ .                                                        |
| $C_{\text{hold}}$                | PV of holding (\$). Societal; discounted at $r_{\text{WACC,real}}$ .                                                            |
| $r_{\text{WACC,real}}$           | Real WACC (decimal). Societal discount rate (§A.2).                                                                             |
| $C_{\text{ROW,capital,nominal}}$ | Total ROW capital costs, nominal (\$).                                                                                          |
| $C_{\text{ROW,capital}}$         | Total ROW capital costs (PV) (\$). Societal perspective.                                                                        |

| Agreement type         | Acquisition                            | Holding                            |
|------------------------|----------------------------------------|------------------------------------|
| New Permanent Easement | Yes ( $\xi_{\text{acquisition}} = 1$ ) | Yes ( $\xi_{\text{holding}} = 1$ ) |
| Simple Fee             | Yes ( $\xi_{\text{acquisition}} = 1$ ) | Yes ( $\xi_{\text{holding}} = 1$ ) |
| Existing Lease/License | No ( $\xi_{\text{acquisition}} = 0$ )  | No ( $\xi_{\text{holding}} = 0$ )  |
| Federal Hybrid         | Yes ( $\xi_{\text{acquisition}} = 1$ ) | Yes ( $\xi_{\text{holding}} = 1$ ) |

## Variables

### Equations

#### 1. Area of ROW per zone.

$$A_{\text{zone}} = \frac{M_{\text{zone}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}}}{43\,560 \text{ (ft}^2\text{/acre)}}.$$

#### 2. Total acquisition cost.

$$C_{\text{acquisition}} = \sum_{\text{zone}} A_{\text{zone}} p_{\text{acquisition,zone}}.$$

#### 3. Annual holding cost.

$$C_{\text{hold,annual}} = \sum_{\text{zone}} A_{\text{zone}} p_{\text{hold,zone}}.$$

Default hold costs (author default) are set to 10% of annual rent per zone, approximating a plausible option/reservation fee ratio for BLM and private ROW agreements.

#### 4. Total holding cost over delay.

$$C_{\text{hold,total}} = C_{\text{hold,annual}} \times T_{\text{delay}}.$$

#### 5. Total nominal ROW capital cost.

$$C_{\text{ROW,capital,nominal}} = \xi_{\text{acquisition}} \cdot C_{\text{acquisition}} + \xi_{\text{holding}} \cdot C_{\text{hold,total}}.$$

**Regulatory perspective (AFUDC).** ROW acquisition is capitalized to COD via S-curve compounding (§A.3); holding costs are not capitalized.

**Societal perspective (present value).** Discount at real WACC:

$$C_{\text{acquisition}} = \frac{C_{\text{acquisition}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}}}}, \quad C_{\text{hold}} = \sum_{t=1}^{T_{\text{delay}}} \frac{C_{\text{hold,annual}}}{(1 + r_{\text{WACC,real}})^t}.$$

Total real (present) value of ROW capital costs:

$$C_{\text{ROW,capital}} = \xi_{\text{acquisition}} \cdot C_{\text{acquisition}} + \xi_{\text{holding}} \cdot C_{\text{hold}}.$$

### B.1.3 Environmental mitigation

The habitat set is the set of habitat terrain types: Forested, Scrubbed Flat, Desert/Barren, Rolling Hills, Mountain. The wetland set is the set of wetland terrain types: Wetland. For reconductoring projects, wetland and habitat credits are zeroed (no ground disturbance). For rebuild projects, credits are scaled by  $f_{\text{rebuild,env}}$  (default 0.5), reflecting partial ground disturbance within an already-cleared corridor. For greenfield, full credits apply. Wetland credits use a single per-acre rate. Habitat credits use per-terrain rates.

| Variable                                               | Description                                                                                                                                                                                                                                                             |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $M_{\text{terrain}}$                                   | Miles of a specific terrain type (miles). Terrain set as in §A.1. Only terrains with $M_{\text{terrain}} > 0$ used.                                                                                                                                                     |
| $W_{\text{ROW}}$                                       | ROW width (ft). Contingent on project category.                                                                                                                                                                                                                         |
| $A_{\text{terrain}}$                                   | Base ROW area in terrain (acres). $A_{\text{terrain}} = (M_{\text{terrain}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}}) / 43\,560 \text{ (ft}^2\text{/acre)}$ .                                                                                                  |
| $u_{\text{mitigation}}$                                | Mitigation uplift factor (dimensionless). Dimensionless (e.g. 1.25). Construction width beyond ROW (TCE).                                                                                                                                                               |
| $A_{\text{terrain}}^{\text{effective}}$                | Effective acres for mitigation in terrain (acres). $A_{\text{terrain}}^{\text{effective}} = A_{\text{terrain}} \times u_{\text{mitigation}}$ .                                                                                                                          |
| $CT$                                                   | Construction type (index). From project category (overhead, underground direct buried, tunnel, subsea).                                                                                                                                                                 |
| $c_{\text{base,peracre}}(\text{terrain}, CT)$          | Base mitigation cost per acre (\$/acre). Depends on terrain and construction type.                                                                                                                                                                                      |
| $C_{\text{base,total}}$                                | Total base mitigation/restoration cost (\$). Sum over terrains ( $M_{\text{terrain}} > 0$ ) of $c_{\text{base,peracre}} \times A_{\text{terrain}}^{\text{effective}}$ .                                                                                                 |
| $r_{\text{marine,env}}$                                | Marine environmental mitigation rate (decimal fraction). Default 0.03 (3% of subsea construction capital expenditure (CAPEX)) (MISO 2025a). Applies to subsea construction type only.                                                                                   |
| $C_{\text{subsea,CAPEX}}$                              | Subsea construction CAPEX (conductor + structure, excl. converter) (\$). From build costs (§B.1.1), terrain-adjusted and contingency-applied.                                                                                                                           |
| $C_{\text{marine,env}}$                                | Marine environmental mitigation cost (\$). $C_{\text{marine,env}} = r_{\text{marine,env}} \times C_{\text{subsea,CAPEX}}$ . Zero when construction type is not subsea.                                                                                                  |
| $A_{\text{wetland}}^{\text{effective impact}}$         | Wetland impact acres (effective) (acres). Wetland terrain only: $A_{\text{wetland}} \times u_{\text{mitigation}}$ . Use only when $M_{\text{terrain}} > 0$ for wetland.                                                                                                 |
| $A_{\text{habitat}}^{\text{effective impact}}$         | Habitat impact acres (effective), total (acres). Sum over habitat terrains $\times u_{\text{mitigation}}$ . Habitat = forested, scrubbed flat, desert barren, rolling hills, mountain. Use only when $M_{\text{terrain}} > 0$ for habitat terrains.                     |
| $A_{\text{terrain,habitat}}^{\text{effective impact}}$ | Effective acres habitat impacted, that terrain (acres). $A_{\text{terrain}} \times u_{\text{mitigation}} \times \mu$ for that terrain. Used in $C_{\text{credits,habitat}}$ . Only when $M_{\text{terrain}} > 0$ for that terrain.                                      |
| $p_{\text{wetland}}$                                   | Wetland credit cost per acre (\$/acre). Default \$51,361/acre (MISO 2025b).                                                                                                                                                                                             |
| $p_{\text{habitat}}(\text{terrain})$                   | Habitat credit cost per acre (\$/acre). Per terrain; explicit value for each habitat terrain.                                                                                                                                                                           |
| $C_{\text{credits,wetlands}}$                          | Wetland credit cost (\$). Use only when $M_{\text{terrain}} > 0$ for wetland.                                                                                                                                                                                           |
| $C_{\text{credits,habitat}}$                           | Habitat credit cost (\$). Sum over habitat terrains of $p_{\text{habitat}}(\text{terrain}) \times A_{\text{terrain,habitat}}^{\text{effective impact}}$ . Use only when $M_{\text{terrain}} > 0$ for habitat terrains.                                                  |
| $\mu$                                                  | Compensatory mitigation ratio (dimensionless, $\geq 1$ ). Required credits per impacted acre. Default 2.0 (modeling assumption for preservation-type mitigation, consistent with the $\geq 1:1$ framework of 33 C.F.R. § 332.3(f) (U.S. Army Corps of Engineers 2008)). |
| $f_{\text{rebuild,env}}$                               | Rebuild environmental mitigation fraction (dimensionless [0,1]). Default 0.5. Only consulted when project type is rebuild. Multiplier on wetland and habitat credit costs; reflects partial ground disturbance within an already-cleared corridor.                      |
| $\xi_{\text{credits}}$                                 | Credit scaling factor (dimensionless [0,1]). 0 for reconductoring; $f_{\text{rebuild,env}}$ for rebuild; 1 for greenfield.                                                                                                                                              |
| $C_{\text{credits,total}}$                             | Total credit cost, wetland + habitat (\$). After project-type scaling.                                                                                                                                                                                                  |
| $T_{\text{delay}}$                                     | Delay years (years).                                                                                                                                                                                                                                                    |
| $T_{\text{construction}}$                              | Construction years (years).                                                                                                                                                                                                                                             |
| $t_{\text{start}}$                                     | Construction start year (year index). $t_{\text{start}} = T_{\text{delay}} + 1$ (1-based).                                                                                                                                                                              |
| $r_{\text{WACC,real}}$                                 | Real WACC (decimal). Societal discount rate (§A.2).                                                                                                                                                                                                                     |
| $C_{\text{env.mit,nominal}}$                           | Total environmental mitigation cost, nominal (\$).                                                                                                                                                                                                                      |
| $B_{\text{env.mit,annual}}$                            | Annual base for env. mitigation (\$/year). Over construction years (PV calculation).                                                                                                                                                                                    |
| $C_{\text{credits}}$                                   | PV of credits (\$).                                                                                                                                                                                                                                                     |
| $C_{\text{env.mit}}$                                   | Total environmental mitigation cost (PV) (\$).                                                                                                                                                                                                                          |

## Variables

## Equations

### 1. Base area per terrain.

$$A_{\text{terrain}} = \frac{M_{\text{terrain}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}}}{43\,560 \text{ (ft}^2\text{/acre)}}.$$

### 2. Effective acres (uplift for TCE).

$$A_{\text{terrain}}^{\text{effective}} = A_{\text{terrain}} \times u_{\text{mitigation}}.$$

### 3. Total base mitigation cost.

$$C_{\text{base,total}} = \sum_{\text{terrain}} c_{\text{base,peracre}}(\text{terrain}, CT) \times A_{\text{terrain}}^{\text{effective}}.$$

**3b. Marine environmental mitigation (subsea only).** Marine environmental mitigation is modeled as a percentage of subsea construction CAPEX rather than a per-acre rate, since no marine mitigation banking market exists and environmental costs for offshore corridors scale with project complexity and value:

$$C_{\text{marine,env}} = r_{\text{marine,env}} \times C_{\text{subsea,CAPEX}}, \quad C_{\text{marine,env}} = 0 \text{ if } CT \neq \text{subsea}.$$

The default  $r_{\text{marine,env}} = 0.03$  is taken from MISO’s 3% engineering, environmental studies, and testing benchmark (MISO 2025a), applied here as a proxy for marine environmental mitigation on subsea CAPEX.  $C_{\text{marine,env}}$  is added to  $C_{\text{base,total}}$  and spread over the construction period on the same schedule as other base mitigation costs.

#### 4. Wetland impact acres (wetland terrain only).

$$A_{\text{wetland}}^{\text{effective impact}} = A_{\text{wetland}} \times u_{\text{mitigation}} \times \mu.$$

#### 5. Habitat impact acres, total.

$$A_{\text{habitat}}^{\text{effective impact}} = \left( \sum_{\text{terrain} \in \text{habitat}} A_{\text{terrain}} \right) \times u_{\text{mitigation}} \times \mu.$$

#### 6. Wetland credit cost.

$$C_{\text{credits,wetlands}} = p_{\text{wetland}} \times A_{\text{wetland}}^{\text{effective impact}}.$$

#### 7. Habitat credit cost.

$$C_{\text{credits,habitat}} = \sum_{\text{terrain} \in \text{habitat}} p_{\text{habitat}}(\text{terrain}) \times A_{\text{terrain,habitat}}^{\text{effective impact}}.$$

#### 8. Total credit cost.

$$C_{\text{credits,total}} = \xi_{\text{credits}} \times (C_{\text{credits,wetlands}} + C_{\text{credits,habitat}}), \quad \text{where } \xi_{\text{credits}} = \begin{cases} 0 & \text{if reconductoring} \\ f_{\text{rebuild,env}} & \text{if rebuild} \\ 1 & \text{if greenfield.} \end{cases}$$

#### 9. Total nominal environmental mitigation cost.

$$C_{\text{env.mit,nominal}} = C_{\text{base,total}} + C_{\text{marine,env}} + C_{\text{credits,total}}.$$

**Regulatory perspective (AFUDC).** Base mitigation and credits are AFUDC-eligible and are capitalized to COD using year-by-year S-curve compounding (see §A.3). Total capitalized environmental mitigation equals capitalized base plus capitalized credits.

**Societal perspective (present value).** Base mitigation spread evenly over  $T_{\text{construction}}$  starting at  $t_{\text{start}}$ . For  $T_{\text{construction}} > 0$ :

$$B_{\text{env.mit,annual}} = \frac{C_{\text{base,total}} + C_{\text{marine,env}}}{T_{\text{construction}}}, \quad C_{\text{base}} = \sum_{t=t_{\text{start}}}^{t_{\text{start}}+T_{\text{construction}}-1} \frac{B_{\text{env.mit,annual}}}{(1 + r_{\text{WACC,real}})^t}.$$

If  $T_{\text{construction}} = 0$ , both base and credits are one-time at  $t_{\text{start}}$ . Otherwise, credits are spread over the construction period (same convention as base costs and AFUDC timing):

$$C_{\text{credits}} = \frac{C_{\text{credits,total}}}{T_{\text{construction}}} \sum_{t=t_{\text{start}}}^{t_{\text{start}}+T_{\text{construction}}-1} \frac{1}{(1 + r_{\text{WACC,real}})^t}, \quad C_{\text{env.mit}} = C_{\text{base}} + C_{\text{credits}}.$$

#### Hard cost total

The three hard-cost components above sum to the hard cost category. Hard costs are the one-time capital expenditures that enter rate base and, from the societal perspective, are discounted from construction spending.

$$C_{\text{hard}} = C_{\text{build}} + C_{\text{ROW,capital}} + C_{\text{env.mit}}.$$

Component PV equations:  $C_{\text{build}}$  (§B.1.1),  $C_{\text{ROW,capital}}$  (§B.1.2),  $C_{\text{env.mit}}$  (§B.1.3).

### B.2 Soft Costs ( $C_{\text{soft}}$ )

Soft costs are ongoing operational, energy-related, and time-value costs. Three sub-aggregates: operational (O&M, insurance, ROW rent), energy (thermal loss), and delay (base delay, congestion delay, displacement delay).

### B.2.1 O&M

| Variable                    | Description                                                                                                                                                                                                                                                                                                                             |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $M_{\text{total}}$          | Total line length (miles). From physical details.                                                                                                                                                                                                                                                                                       |
| $M_{\text{terrain}}$        | Miles in terrain type (miles). Terrain set as in §A.1. Only terrains with $M_{\text{terrain}} > 0$ used in sums.                                                                                                                                                                                                                        |
| $T_{\text{delay}}$          | Delay years (years). Before construction.                                                                                                                                                                                                                                                                                               |
| $T_{\text{construction}}$   | Construction years (years).                                                                                                                                                                                                                                                                                                             |
| $T_{\text{lifetime}}$       | Project lifetime (years). O&M accrues over this.                                                                                                                                                                                                                                                                                        |
| $T_{\text{COD}}$            | Commercial operation date (year index). $T_{\text{COD}} = T_{\text{delay}} + T_{\text{construction}} + 1$ (1-based; first year of O&M).                                                                                                                                                                                                 |
| $r_{\text{WACC,real}}$      | Real WACC (decimal). Used to discount O&M (§A.2).                                                                                                                                                                                                                                                                                       |
| $CT$                        | Construction type (index). Overhead, Underground direct-buried, Underground tunnel, Subsea.                                                                                                                                                                                                                                             |
| $c_{\text{base}}$           | Base line O&M rate (\$/mile/year). Overhead only. \$19,090/mi/yr, 2025\$: CT Siting Council (2022) \$29,636 (2017–2021 mean, ~2019\$) escalated to 2025\$ minus vegetation management (2025\$).                                                                                                                                         |
| $r_{\text{converter,O\&M}}$ | Converter O&M rate (decimal fraction). Annual O&M as fraction of converter station CAPEX. 0.005 (0.5% of station CAPEX/yr) for all HVDC converter types (Universidad de Castilla-La Mancha 2023).                                                                                                                                       |
| $K_{\text{converter}}$      | Total converter station CAPEX (\$). Equal to the fixed converter-station capital cost component of build cost (§B.1.1).                                                                                                                                                                                                                 |
| $r_{\text{line,O\&M}}$      | Line O&M rate (decimal fraction). Non-overhead only. Annual O&M as fraction of line CAPEX. UG direct-buried: 0.0015 (Centre for Infrastructure Engineering and Technology (CIET) and Powerlink Queensland 2023); tunnel: 0.004 (Institution of Engineering and Technology and Mott MacDonald 2025); subsea: 0.025 (NeoMarketData 2025). |
| $K_{\text{line}}$           | Line CAPEX (excl. converters) (\$). $K_{\text{line}} = C_{\text{conductor,build}} + C_{\text{structure,build}}$ from §B.1.1.                                                                                                                                                                                                            |
| $\xi_{\text{DC}}$           | Binary for DC projects (0 or 1). 1 if DC, 0 if AC. Gates converter O&M (§B.1.1).                                                                                                                                                                                                                                                        |
| $c_{\text{veg(terrain)}}$   | Vegetation management (\$/mile/year). By terrain. Overhead only.                                                                                                                                                                                                                                                                        |
| $C_{\text{O\&M,annual}}$    | Total annual O&M (\$/year). $C_{\text{line,annual}} + C_{\text{converter,annual}}$ .                                                                                                                                                                                                                                                    |
| $C_{\text{O\&M,nominal}}$   | O&M cost (nominal) (\$). Undiscounted lifetime O&M.                                                                                                                                                                                                                                                                                     |
| $g_{\text{O\&M}}$           | Real O&M escalation rate (decimal). Default 2.0% real/yr (Connecticut Siting Council 2023). Applied via growing annuity, same formula as wildfire (§B.3.1) and outage (§B.3.2) escalation.                                                                                                                                              |
| $C_{\text{O\&M}}$           | O&M cost (PV) (\$). Present value of O&M from $T_{\text{COD}}$ over $T_{\text{lifetime}}$ .                                                                                                                                                                                                                                             |

### Variables

### Equations

#### 1. COD (first year of O&M).

$$T_{\text{COD}} = T_{\text{delay}} + T_{\text{construction}} + 1.$$

#### 2. Line O&M (annual).

$$C_{\text{line,annual}} = \begin{cases} c_{\text{base}} \times M_{\text{total}} + C_{\text{veg,annual}} & \text{if overhead} \\ r_{\text{line,O\&M}} \times K_{\text{line}} & \text{if non-overhead} \end{cases}$$

where:

$$C_{\text{veg,annual}} = \sum_{\text{terrain}} M_{\text{terrain}} \times c_{\text{veg(terrain)}}.$$

The overhead base rate  $c_{\text{base}} = \$19,090/\text{circuit-mile/year}$  (2025\$) is derived from Connecticut Siting Council total O&M (Connecticut Siting Council 2023), escalated to 2025\$, net of separately modeled vegetation management. This single rate covers conductor inspection, structure maintenance, patrol, and all non-vegetation line operating costs.

For non-overhead construction types, O&M scales with installed asset value rather than route length. Published benchmarks report total O&M per circuit-mile or as a percentage of capital cost, without a conductor/structure decomposition for underground or subsea systems. Rates: underground direct-buried 0.15% (Centre for Infrastructure Engineering and Technology (CIET) and Powerlink Queensland 2023), consistent with Connecticut Siting Council FERC Form 1 data (\$22,937/circuit-mile; Connecticut Siting Council 2023); tunnel 0.4% (Institution of Engineering and Technology and Mott MacDonald 2025); subsea 2.5% (NeoMarketData 2025), an industry estimate for cable-specific maintenance excluding whole-link operational costs (insurance, staffing, trading).

**Vegetation management defaults (overhead).** Vegetation management costs  $c_{\text{veg(terrain)}}$  reflect U.S. utility spending data by terrain archetype, expressed in 2025\$ via CPI-U escalation from source years. Default values are calibrated to regulatory filings and industry publications: forested terrain (\$14,367/mi/yr, author estimate); mountain terrain (\$12,314/mi/yr) reflecting helicopter access premiums of 5.7–8× ground crew costs (Bonneville Power Administration 2021; LIFE Elia-RTE 2015); wetland (\$0/mi/yr); scrubbed/brush (\$5,131/mi/yr) from

mechanical mulching costs below forest clearing (Backwell 2024); rolling hills (\$4,747/mi/yr, 2025\$, from Duke Energy Indiana transmission VM average (Duke Energy Indiana, LLC 2023)); farmland (\$1,196/mi/yr) and desert/barren (\$449/mi/yr) author estimates; urban (\$1,000/mi/yr, author judgment: overhead transmission through dense urban cores is structurally rare and minimal vegetation exists where it does; utility sources aggregate urban transmission VM with broader corridor data). Defaults may be varied for project-specific conditions including wildfire risk zone and local growth rates. Non-overhead construction types have zero vegetation management cost.

### 3. Converter (annual; DC only).

$$C_{\text{converter,annual}} = \xi_{\text{DC}} \times r_{\text{converter,O\&M}} \times K_{\text{converter}}.$$

Converter O&M scales with station capacity, not line length. The rate is 0.5% of converter station CAPEX per year for all HVDC converter types (Universidad de Castilla-La Mancha 2023).

### 4. Total annual O&M.

$$C_{\text{O\&M,annual}} = C_{\text{line,annual}} + C_{\text{converter,annual}}.$$

**5. Nominal (undiscounted) lifetime O&M.** O&M costs escalate at real rate  $g_{\text{O\&M}}$  per year:

$$C_{\text{O\&M,nominal}} = \sum_{t=0}^{T_{\text{lifetime}}-1} C_{\text{O\&M,annual}} \times (1 + g_{\text{O\&M}})^t.$$

**6. Real (present value) O&M.** O&M costs escalate at real rate  $g_{\text{O\&M}}$  from  $T_{\text{COD}}$ , using the same growing-annuity PV as in §B.3.1:

$$C_{\text{O\&M}} = \frac{C_{\text{O\&M,annual}} \times \frac{1 - \left( \frac{1 + g_{\text{O\&M}}}{1 + r_{\text{WACC,real}}} \right)^{T_{\text{lifetime}}}}{r_{\text{WACC,real}} - g_{\text{O\&M}}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

When  $g_{\text{O\&M}} = 0$ , this reduces to the flat-annuity form. The default  $g_{\text{O\&M}} = 2.0\%$  real/yr is the Connecticut Siting Council's official 40-year lifecycle compound annual growth rate for transmission O&M (Connecticut Siting Council 2023), directionally consistent with FERC Form 1 analyses showing rising real transmission costs for U.S. investor-owned utilities driven by infrastructure investment in the 2010s (Fares and King 2017).

## B.2.2 Operational insurance

Overhead and underground lines are predominantly self-insured (commercial property insurance for T&D lines is generally unavailable); subsea cables and converter stations are commercially insured.

| Variable                        | Description                                                                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $C_{\text{adj,cont,conductor}}$ | Conductor build cost (\$). Terrain-adjusted, contingency applied. From Build costs (§B.1.1).                                                                |
| $C_{\text{adj,cont,structure}}$ | Structure build cost (\$). Terrain-adjusted, contingency applied. From Build costs (§B.1.1).                                                                |
| $C_{\text{adj,cont,converter}}$ | Converter build cost (\$). Terrain-adjusted, contingency applied. From Build costs (§B.1.1).                                                                |
| $\xi_{\text{DC}}$               | Binary for DC projects (0 or 1). 1 if DC, 0 if AC. Gates converter in insurable value (§B.1.1).                                                             |
| $\xi_{\text{ins,conductor}}$    | Include conductors in insurable value (0 or 1). 1 if insured, 0 if not.                                                                                     |
| $\xi_{\text{ins,structure}}$    | Include structures in insurable value (0 or 1). 1 if insured, 0 if not.                                                                                     |
| $\xi_{\text{ins,converter}}$    | Include converters in insurable value (0 or 1). 1 if insured, 0 if not.                                                                                     |
| $V_{\text{insurable}}$          | Insurable asset value (\$). Sum of included build-cost components (with $\xi_{\text{DC}}$ for converter).                                                   |
| $CT$                            | Construction type (index). Overhead, Underground direct-buried, Underground tunnel, Subsea.                                                                 |
| $r_{\text{premium}}(CT)$        | Premium rate (decimal). Annual premium as fraction of insurable value. Can depend on $CT$ .                                                                 |
| $r_{\text{premium,default}}$    | Default premium rate (decimal). Used when no type-specific rate is defined for $CT$ .                                                                       |
| $C_{\text{insurance,annual}}$   | Annual operational insurance premium (\$/year).                                                                                                             |
| $T_{\text{lifetime}}$           | Project lifetime (years).                                                                                                                                   |
| $T_{\text{COD}}$                | Commercial operation date (year index). First year premium is paid. $T_{\text{COD}} = T_{\text{delay}} + T_{\text{construction}} + 1$ .                     |
| $r_{\text{WACC,real}}$          | Real WACC (decimal). Used to discount premiums (§A.2).                                                                                                      |
| $C_{\text{insurance,nominal}}$  | Operational insurance cost (nominal) (\$). Undiscounted sum of premiums over project lifetime.                                                              |
| $g_{\text{ins}}$                | Insurance escalation rate (decimal). Default 2.0% real/yr. Author assumption reflecting long-term climate risk trends; short-term market cycles may differ. |
| $C_{\text{insurance}}$          | Operational insurance cost (PV) (\$). Present value of premium stream from $T_{\text{COD}}$ over $T_{\text{lifetime}}$ .                                    |

## Variables

## Equations

### 1. Insurable asset value.

$$V_{\text{insurable}} = \xi_{\text{ins,conductor}} \cdot C_{\text{adj,cont,conductor}} + \xi_{\text{ins,structure}} \cdot C_{\text{adj,cont,structure}} + \xi_{\text{ins,converter}} \cdot \xi_{\text{DC}} \cdot C_{\text{adj,cont,converter}}$$

The insurable value includes contingency and soft-cost multiplier applied upstream in §B.1.1.

### 2. Premium rate.

$$r_{\text{premium}}(CT) = \begin{cases} r_{\text{premium},CT} & \text{if a rate is specified for construction type } CT \\ r_{\text{premium,default}} & \text{otherwise} \end{cases}$$

**3. Annual premium.** The premium is computed separately for line assets and converter stations, since converter stations carry higher commercial rates:

$$C_{\text{insurance,annual}} = V_{\text{insurable,line}} \times r_{\text{premium,line}}(CT) + V_{\text{insurable,conv}} \times r_{\text{premium,conv}},$$

where  $V_{\text{insurable,line}} = \xi_{\text{ins,conductor}} \cdot C_{\text{adj,cont,conductor}} + \xi_{\text{ins,structure}} \cdot C_{\text{adj,cont,structure}}$  and  $V_{\text{insurable,conv}} = \xi_{\text{ins,converter}} \cdot \xi_{\text{DC}} \cdot C_{\text{adj,cont,converter}}$ .

### 4. Nominal (undiscounted) lifetime cost.

$$C_{\text{insurance,nominal}} = C_{\text{insurance,annual}} \times \frac{(1 + g_{\text{ins}})^{T_{\text{lifetime}}} - 1}{g_{\text{ins}}}.$$

**5. Real (present value) cost.** Risk-bearing costs escalate at real rate  $g_{\text{ins}}$  from  $T_{\text{COD}}$ :

$$C_{\text{insurance}} = \frac{C_{\text{insurance,annual}} \times \frac{1 - \left( \frac{1 + g_{\text{ins}}}{1 + r_{\text{WACC,real}}} \right)^{T_{\text{lifetime}}}}{r_{\text{WACC,real}} - g_{\text{ins}}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

#### Default calibration

| Construction type           | $r_{\text{premium}}(CT)$ | Basis                                                                                     |
|-----------------------------|--------------------------|-------------------------------------------------------------------------------------------|
| Overhead                    | 0.20%                    | Author default; self-insurance reserve, consistent with utility practice                  |
| Underground (direct-buried) | 0.20%                    | Same mechanism                                                                            |
| Underground (tunnel)        | 0.20%                    | Same mechanism                                                                            |
| Subsea                      | 3.0%                     | Author default; reflects commercial property-damage premiums for submarine infrastructure |
| Converter                   | 0.70%                    | Author default; reflects commercial premiums for power-electronics assets                 |

### B.2.3 Operational ROW rent

Default per-acre zone values and annual rents from the BLM Per-Acre Linear Rent Schedule for Calendar Years 2026–2035 (Bureau of Land Management 2026b); these defaults apply to federal (BLM) lands and private-land projects should substitute site-specific values. Same zones, areas, and per-acre rent as in Capital ROW (§B.1.2); agreement type determines whether rent applies and over which years.

| Variable                        | Description                                                                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| zone                            | Zone index (index). Same zones as Capital ROW; use only zones where $M_{\text{zone}} > 0$ .                                                                 |
| $M_{\text{zone}}$               | Miles in zone (miles).                                                                                                                                      |
| $W_{\text{ROW}}$                | ROW width (ft). Same as Capital ROW (project category).                                                                                                     |
| $A_{\text{zone}}$               | Area of ROW of zone (acres). $A_{\text{zone}} = (M_{\text{zone}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}}) / 43\,560 \text{ (ft}^2\text{/acre)}$ . |
| $p_{\text{rent,zone}}$          | Rent per acre per year, zone (\$/acre/year).                                                                                                                |
| $C_{\text{rent,annual}}$        | Total annual ROW rent (\$/year). Sum over zones.                                                                                                            |
| $\xi_{\text{rent}}$             | Binary: rent applies (0 or 1). 1 for Existing Lease/License or Federal Hybrid; 0 for New Permanent Easement or Simple Fee.                                  |
| $T_{\text{delay}}$              | Delay years (years).                                                                                                                                        |
| $T_{\text{construction}}$       | Construction years (years).                                                                                                                                 |
| $T_{\text{lifetime}}$           | Project lifetime (years). Years of operation.                                                                                                               |
| $t_{\text{rent,start}}$         | First year rent is paid (year index). Agreement-dependent (see table).                                                                                      |
| $T_{\text{rent}}$               | Number of years rent is paid (years). Agreement-dependent (see table).                                                                                      |
| $g_{\text{rent}}$               | ROW rent escalation rate (decimal). Real annual escalation applied to annual rent. Default 0% real (§A.2).                                                  |
| $r_{\text{WACC,real}}$          | Real WACC (decimal). Used to discount rent (§A.2).                                                                                                          |
| $C_{\text{rent,total,nominal}}$ | ROW rent cost (nominal) (\$). Undiscounted sum over all years rent is paid.                                                                                 |
| $C_{\text{rent,total}}$         | ROW rent cost (PV) (\$). Present value of rent stream.                                                                                                      |

| Agreement type         | Rent?                           | $t_{\text{rent,start}}$ | $T_{\text{rent}}$                                                  |
|------------------------|---------------------------------|-------------------------|--------------------------------------------------------------------|
| New Permanent Easement | No ( $\xi_{\text{rent}} = 0$ )  | —                       | —                                                                  |
| Simple Fee             | No ( $\xi_{\text{rent}} = 0$ )  | —                       | —                                                                  |
| Existing Lease/License | Yes ( $\xi_{\text{rent}} = 1$ ) | 1                       | $T_{\text{delay}} + T_{\text{construction}} + T_{\text{lifetime}}$ |
| Federal Hybrid         | Yes ( $\xi_{\text{rent}} = 1$ ) | $T_{\text{delay}} + 1$  | $T_{\text{construction}} + T_{\text{lifetime}}$                    |

## Variables

### Equations

#### 1. Area of ROW per zone.

$$A_{\text{zone}} = \frac{M_{\text{zone}} \times 5280 \text{ (ft/mi)} \times W_{\text{ROW}}}{43\,560 \text{ (ft}^2\text{/acre)}}.$$

#### 2. Annual ROW rent.

$$C_{\text{rent,annual}} = \sum_{\text{zone}} A_{\text{zone}} p_{\text{rent,zone}}.$$

#### 3. Nominal (undiscounted) ROW rent cost. Growing annual rent at rate $g_{\text{rent}}$ over $T_{\text{rent}}$ years:

$$C_{\text{rent,total,nominal}} = \xi_{\text{rent}} \times C_{\text{rent,annual}} \times \frac{(1 + g_{\text{rent}})^{T_{\text{rent}}} - 1}{g_{\text{rent}}}.$$

At  $g_{\text{rent}} = 0$  this reduces to  $\xi_{\text{rent}} \times C_{\text{rent,annual}} \times T_{\text{rent}}$ .

#### 4. Real (present value) ROW rent cost. Growing annuity from $t_{\text{rent,start}}$ for $T_{\text{rent}}$ years, discounted at $r_{\text{WACC,real}}$ :

$$C_{\text{rent,total}} = \frac{\xi_{\text{rent}} \times C_{\text{rent,annual}} \times \frac{1 - \left(\frac{1 + g_{\text{rent}}}{1 + r_{\text{WACC,real}}}\right)^{T_{\text{rent}}}}{r_{\text{WACC,real}} - g_{\text{rent}}}}{(1 + r_{\text{WACC,real}})^{t_{\text{rent,start}} - 1}}.$$

At  $g_{\text{rent}} = 0$  this reduces to the flat-annuity form. The structure matches the growing-annuity PV in §B.3.1.

The default  $g_{\text{rent}} = 0$  (0% real) reflects that federal BLM rents adjust at the Implicit Price Deflator for GDP rate (2.6%/yr nominal for CY 2026–2035), approximately matching inflation (Bureau of Land Management 2026a; U.S. Government 2024). State and private ROW leases similarly escalate at 2.5–3%/yr nominal—no evidence supports systematic above-inflation rent growth for transmission corridors. A positive  $g_{\text{rent}}$  may be examined in sensitivity analysis for urban corridor appreciation scenarios.

## B.2.4 Energy losses

| Variable                             | Description                                                                                                                                                                          |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $C_{\text{new}}$                     | New line capacity (MW). Nameplate. All project types.                                                                                                                                |
| $V$                                  | Line voltage (kV). Reconductoring: existing (old) config (towers unchanged); greenfield and rebuild: new config.                                                                     |
| $L$                                  | Line length (miles). Total line length.                                                                                                                                              |
| $u$                                  | Line utilization (dimensionless). $0 \leq u \leq 1$ . Average fraction of capacity used.                                                                                             |
| $n_{\text{phases}}$                  | Number of phases (count). 3 for AC; 2 for DC (poles).                                                                                                                                |
| $n_{\text{circuits}}$                | Number of circuits / poles (count). Parallel circuits (AC) or poles (DC).                                                                                                            |
| $n_{\text{conductorsperphase}}$      | Conductors per phase (or per pole) (count). Conductors in parallel per phase/pole.                                                                                                   |
| $R_{\text{mi}}$                      | Resistance per mile ( $\Omega/\text{mile}$ ). AC: 75°C; DC: corrected to operating temperature. Depends on conductor and circuit.                                                    |
| $\alpha_{20}$                        | Temperature coefficient of resistance at 20°C ( $\text{K}^{-1}$ ). DC only. Aluminum: 0.00403; copper: 0.00393 (IEC 2023a).                                                          |
| $T_{\text{DC,op}}$                   | DC operating temperature (°C). Default 75°C, matching the AC 75°C convention (IEEE 2023; IEC 2023b).                                                                                 |
| $I$                                  | Phase current (A). RMS phase current at full capacity and nominal voltage.                                                                                                           |
| $\phi_{\text{AC}}$                   | AC power factor (dimensionless). e.g. 0.95. AC only.                                                                                                                                 |
| $f_{\text{load}}$                    | Full-load adjustment factor (dimensionless). $0.3u + 0.7u^2$ ; empirical coefficients for North American transmission (NaitBelaid, Heleno, and LaCommare 2024).                      |
| $H$                                  | Hours per year (h/year). e.g. 8760.                                                                                                                                                  |
| $T_{\text{lifetime}}$                | Project lifetime (years).                                                                                                                                                            |
| $\xi_{\text{DC}}$                    | Binary for DC (0 or 1). 1 if DC, 0 if AC.                                                                                                                                            |
| $N_{\text{conv}}$                    | Number of converter stations (count). 0 for AC; typically 2 for DC.                                                                                                                  |
| $\lambda_{\text{conv}}$              | Converter loss fraction (dimensionless). Fraction of through-power lost per station (e.g. LCC, VSC).                                                                                 |
| $P_{\text{lineloss,MW}}$             | Line loss (power) (MW). Total resistive line loss (average power).                                                                                                                   |
| $E_{\text{lineloss,MW,annual}}$      | Line energy losses (annual) (MWh/year).                                                                                                                                              |
| $E_{\text{lineloss,lifetime}}$       | Line energy losses (lifetime) (MWh).                                                                                                                                                 |
| $P_{\text{corona,MW}}$               | Corona loss (power) (MW). AC overhead only; voltage-dependent, load-independent. Zero for DC and non-overhead.                                                                       |
| $c_{\text{corona}}(V)$               | Corona loss by voltage class (kW/mile). 0 ( $\leq 230$ kV), 6 (345 kV), 16 (500 kV), 44 (765 kV, midpoint of 23–64 range); annual average, sea level (American Electric Power n.d.). |
| $P_{\text{converterloss,MW}}$        | Converter loss (power) (MW). DC only.                                                                                                                                                |
| $E_{\text{converterloss,MW,annual}}$ | Converter energy losses (annual) (MWh/year).                                                                                                                                         |
| $E_{\text{loss}}$                    | Total energy losses (annual) (MWh/year). Line + converter.                                                                                                                           |
| $E_{\text{totalloss,lifetime}}$      | Total energy losses (lifetime) (MWh).                                                                                                                                                |

## Variables

## Equations

### 1. Phase current (from capacity and voltage).

$$I = \frac{C_{\text{new}} \times 1000 \text{ (A}\cdot\text{kV/MW)}}{\phi_{\text{AC}} \times V \times \sqrt{n_{\text{phases}}} \times n_{\text{circuits}}} \quad (\text{AC}), \quad I = \frac{C_{\text{new}} \times 1000 \text{ (A}\cdot\text{kV/MW)}}{V \times \sqrt{n_{\text{phases}}} \times n_{\text{circuits}}} \quad (\text{DC}).$$

### 2. Full-load adjustment (utilization).

$$f_{\text{load}} = 0.3u + 0.7u^2.$$

**3. DC resistance temperature correction (DC only).** DC resistance is tabulated at 20°C reference temperature but conductors operate at elevated temperature under load. Per IEC 60287-1-1:2023 and IEEE Std 738-2023, resistance is corrected linearly:

$$R_{\text{mi}} = R_{\text{DC},20} \times [1 + \alpha_{20}(T_{\text{DC,op}} - 20)] \quad (\text{DC only}).$$

At the default  $T_{\text{DC,op}} = 75^\circ\text{C}$ , this increases DC resistance by 22.2% for aluminum ( $\alpha_{20} = 0.00403$ ) and 21.6% for copper ( $\alpha_{20} = 0.00393$ ) relative to the uncorrected 20°C value (IEC 2023a). AC resistance ( $R_{\text{mi}} = R_{\text{AC},75}$ ) is already tabulated at 75°C and requires no correction.

**4. Line loss (power).** Total resistive ( $I^2R$ ) line loss, average power in MW:

$$P_{\text{lineloss,MW}} = \left( \frac{I}{n_{\text{conductorsperphase}}} \right)^2 \times (n_{\text{conductorsperphase}} \times n_{\text{phases}} \times n_{\text{circuits}}) \times R_{\text{mi}} \times L \times \frac{f_{\text{load}}}{10^6}.$$

### 5. Line energy losses (annual and lifetime).

$$E_{\text{lineloss,MW,annual}} = P_{\text{lineloss,MW}} \times H, \quad E_{\text{lineloss,lifetime}} = E_{\text{lineloss,MW,annual}} \times T_{\text{lifetime}}.$$

**6. Corona loss (AC overhead only).** Corona loss occurs when conductor-surface electric field strength exceeds the dielectric breakdown gradient of air; it is voltage-dependent and load-independent, so it is modeled as an additive term rather than a multiplier on  $I^2R$  losses:

$$P_{\text{corona,MW}} = \xi_{\text{AC}} \times \xi_{\text{overhead}} \times \frac{c_{\text{corona}}(V) \times L \times n_{\text{circuits}}}{1000}.$$

Corona is added to total line loss before computing annual and lifetime energy losses:  $P_{\text{lineloss,MW}} += P_{\text{corona,MW}}$ . Values are annual averages at sea level, weighted by a typical 20%/2%/78% rain/snow/fair-weather mix (American Electric Power n.d.); higher altitude increases corona by approximately 25% per 1,000 m of elevation, and foul-weather corona can exceed fair-weather corona by 20:1. DC corona exists but is much smaller than AC corona and is excluded.

## 7. Converter loss (DC only).

$$P_{\text{converterloss,MW}} = \xi_{\text{DC}} \times N_{\text{conv}} \times \lambda_{\text{conv}} \times u \times C_{\text{new}}, \quad E_{\text{converterloss,MW,annual}} = P_{\text{converterloss,MW}} \times H.$$

Default converter loss fractions: LCC 0.75%, VSC 1.0%. The VSC default reflects a midpoint of the published range: ACORE reports 0.7% for modular multi-level converters (ACORE 2023), while broader VSC survey data spans 1.2–1.5% (Akhatov et al. 2025).

## 8. Total energy losses.

$$E_{\text{loss}} = E_{\text{lineloss,MW,annual}} + E_{\text{converterloss,MW,annual}}, \quad E_{\text{totalloss,lifetime}} = E_{\text{loss}} \times T_{\text{lifetime}}.$$

### B.2.5 Thermal loss costs

Per-conductor AC resistance at 75°C is sourced from the Southwire Overhead Conductor Manual (2018) for ACSR configurations, CTC Global published datasheets for ACCC, and IEC 60228/60287 for underground XLPE cables.

| Variable                             | Description                                                                                                                                                                                           |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_{\text{loss}}$                    | Total energy losses (annual) (MWh/year). From §B.2.4 (line + converter).                                                                                                                              |
| $E_{\text{lineloss,MW,annual}}$      | Line energy losses (annual) (MWh/year). Conductor only.                                                                                                                                               |
| $E_{\text{converterloss,MW,annual}}$ | Converter energy losses (annual) (MWh/year). DC only.                                                                                                                                                 |
| $v_{\text{load}}$                    | Value of load (\$/MWh). Marginal value of energy to end-use load. The value of electricity derives from the services load performs with it, making demand-side valuation the correct welfare measure. |
| $T_{\text{lifetime}}$                | Project lifetime (years).                                                                                                                                                                             |
| $T_{\text{COD}}$                     | Commercial operation date (year). First year of operation.                                                                                                                                            |
| $r_{\text{WACC,real}}$               | Real WACC (decimal). Used to discount this cost (§A.2).                                                                                                                                               |
| $C_{\text{lineloss,annual}}$         | Line loss cost (annual) (\$/year). Cost of line losses only.                                                                                                                                          |
| $C_{\text{converterloss,annual}}$    | Converter loss cost (annual) (\$/year). Cost of converter losses only.                                                                                                                                |
| $C_{\text{loss,annual}}$             | Total thermal loss cost (annual) (\$/year). Line + converter.                                                                                                                                         |
| $g_{\text{benefit}}$                 | Benefit price escalation rate (decimal). From §A.2. Applied to energy valuation for welfare consistency.                                                                                              |
| $C_{\text{loss,nominal}}$            | Total thermal loss cost (nominal) (\$). Undiscounted sum over lifetime.                                                                                                                               |
| $C_{\text{loss}}$                    | Total thermal loss cost (PV) (\$). PV of growing annual stream from $T_{\text{COD}}$ over $T_{\text{lifetime}}$ .                                                                                     |

## Variables

## Equations

### 1. Component costs (line and converter).

$$C_{\text{lineloss,annual}} = E_{\text{lineloss,MW,annual}} \times v_{\text{load}}, \quad C_{\text{converterloss,annual}} = E_{\text{converterloss,MW,annual}} \times v_{\text{load}}.$$

### 2. Total annual thermal loss cost.

$$C_{\text{loss,annual}} = E_{\text{loss}} \times v_{\text{load}} = C_{\text{lineloss,annual}} + C_{\text{converterloss,annual}}.$$

### 3. Nominal (undiscounted) lifetime cost. Growing annual loss cost at rate $g_{\text{benefit}}$ over $T_{\text{lifetime}}$ years:

$$C_{\text{loss,nominal}} = C_{\text{loss,annual}} \times \frac{(1 + g_{\text{benefit}})^{T_{\text{lifetime}}} - 1}{g_{\text{benefit}}}.$$

At  $g_{\text{benefit}} = 0$  this reduces to  $C_{\text{loss,annual}} \times T_{\text{lifetime}}$ .

**4. Real (present value) cost.** Growing annuity from  $T_{\text{COD}}$  for  $T_{\text{lifetime}}$  years, discounted at  $r_{\text{WACC,real}}$ :

$$C_{\text{loss}} = \frac{C_{\text{loss,annual}} \times \frac{1 - \left( \frac{1+g_{\text{benefit}}}{1+r_{\text{WACC,real}}} \right)^{T_{\text{lifetime}}}}{r_{\text{WACC,real}} - g_{\text{benefit}}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

At  $g_{\text{benefit}} = 0$  this reduces to the flat-annuity form. The structure matches the growing-annuity PV in §B.3.1.

The value of a MWh lost on the line is the same commodity as a MWh delivered to load ( $v_{\text{load}}$ , §C.2.1). Welfare consistency requires that both escalate at  $g_{\text{benefit}}$ . The loss-compensation emissions cost ( $C_{\text{comp,emissions}}$ ) is separate and escalates at  $g_{c,k}$  (SCC growth, §B.4).

### Energy loss cost aggregate

Congestion relief (§A.4) covers constrained-hour redispatch savings, so there is no separate residual-exceedance cost; the energy cost aggregate is thermal loss cost only:

$$C_{\text{energy}} = C_{\text{loss}}.$$

### B.2.6 Base delay costs

Published sources lack component-level benchmark rates for these costs; the rates are project-specific. Foregone congestion benefits during delay are modeled separately (§B.2.7).

| Variable                   | Description                                                                                                             |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| $C_{\text{delay,annual}}$  | Annual base delay cost (\$/year). Total direct out-of-pocket annual spend during delay. Exogenous parameter; default 0. |
| $T_{\text{delay}}$         | Delay years (years). From project technical details; permitting/pre-construction period.                                |
| $r_{\text{WACC,real}}$     | Real WACC (decimal). Used to discount the delay cost stream (§A.2).                                                     |
| $C_{\text{delay,nominal}}$ | Total base delay cost (nominal) (\$). $C_{\text{delay,nominal}} = C_{\text{delay,annual}} \times T_{\text{delay}}$ .    |
| $C_{\text{base,delay}}$    | Total base delay cost (PV) (\$). PV of level annuity over $T_{\text{delay}}$ years at $r_{\text{WACC,real}}$ .          |

### Variables

#### Equations

**1. Annual base delay cost.**  $C_{\text{delay,annual}}$  is specified directly (no decomposition).

**2. Nominal total base delay cost.**

$$C_{\text{delay,nominal}} = C_{\text{delay,annual}} \times T_{\text{delay}}.$$

**3. Real (present value) base delay cost.** Level annuity over years  $t = 1, \dots, T_{\text{delay}}$ , discounted at  $r_{\text{WACC,real}}$ :

$$C_{\text{base,delay}} = C_{\text{delay,annual}} \times \frac{1 - (1 + r_{\text{WACC,real}})^{-T_{\text{delay}}}}{r_{\text{WACC,real}}}.$$

See §B.2.7 for congestion delay costs. Delay cost categories apply only pre-COD; benefit streams in §C apply only post-COD. No year falls in both periods, so there is no double-count.

### B.2.7 Congestion delay costs

Congestion delay costs are the opportunity cost of foregone congestion relief during delay and construction. The annual cost equals  $B_{\text{congestion,annual}}$  (§C), growing at  $g_{\text{benefit}}$  over  $T_{\text{delay}} + T_{\text{construction}}$  years.

| Variable                        | Description                                                                                                                                                    |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $B_{\text{congestion,annual}}$  | Annual congestion reduction benefit (\$/year). Same formula as in §C: congestion relief MWh/year $\times \gamma_{\text{congestion}}$ .                         |
| $C_{\text{cong,delay,annual}}$  | Congestion delay cost (annual) (\$/year). $C_{\text{cong,delay,annual}} = B_{\text{congestion,annual}}$ .                                                      |
| $T_{\text{delay}}$              | Delay years. From project technical details.                                                                                                                   |
| $T_{\text{construction}}$       | Construction years. From project technical details.                                                                                                            |
| $r_{\text{WACC,real}}$          | Real WACC (decimal). Used to discount the delay cost stream (§A.2).                                                                                            |
| $g_{\text{benefit}}$            | Benefit price escalation rate (decimal). From §A.2. The same commodity price growth that applies to the post-COD congestion benefit.                           |
| $C_{\text{cong,delay,nominal}}$ | Congestion delay cost (nominal) (\$). Total over delay + construction; undiscounted.                                                                           |
| $C_{\text{cong,delay}}$         | Congestion delay cost (PV) (\$). PV of growing annuity over $T_{\text{delay}} + T_{\text{construction}}$ years at $r_{\text{WACC,real}}$ , starting at year 1. |

## Variables

### Equations

**1. Annual congestion delay cost.** The annual opportunity cost equals the annual congestion reduction benefit (congestion relief  $\times \gamma_{\text{congestion}}$ ; see §C):

$$C_{\text{cong,delay,annual}} = B_{\text{congestion,annual}}.$$

**2. Nominal total congestion delay cost.** Growing annual cost at rate  $g_{\text{benefit}}$  over  $T_{\text{delay}} + T_{\text{construction}}$  years:

$$C_{\text{cong,delay,nominal}} = C_{\text{cong,delay,annual}} \times \frac{(1 + g_{\text{benefit}})^{T_{\text{delay}} + T_{\text{construction}}} - 1}{g_{\text{benefit}}}.$$

At  $g_{\text{benefit}} = 0$  this reduces to  $C_{\text{cong,delay,annual}} \times (T_{\text{delay}} + T_{\text{construction}})$ .

**3. Real (present value) congestion delay cost.** Growing annuity over years  $t = 1, \dots, T_{\text{delay}} + T_{\text{construction}}$ , discounted at  $r_{\text{WACC,real}}$ :

$$C_{\text{cong,delay}} = C_{\text{cong,delay,annual}} \times \frac{1 - \left( \frac{1 + g_{\text{benefit}}}{1 + r_{\text{WACC,real}}} \right)^{T_{\text{delay}} + T_{\text{construction}}}}{r_{\text{WACC,real}} - g_{\text{benefit}}}.$$

At  $g_{\text{benefit}} = 0$  this reduces to the flat-annuity form.

*Link: Annual congestion delay cost uses the same  $B_{\text{congestion,annual}}$  as the post-COD benefit (§C); pre-COD delay and post-COD benefit periods are disjoint, so no double-count. Base delay (§B.2.6) and displacement delay (§C.3.1) use the same period and discounting.*

### Soft cost total

The soft-cost components above sum to the soft cost category.  $C_{\text{emissions}}$  is a separate category (§B.4); see also §D. Soft costs are ongoing operational, energy-related, and time-value costs.

$$\begin{aligned} C_{\text{operational}} &= C_{\text{O\&M}} + C_{\text{insurance}} + C_{\text{rent,total}}, \\ C_{\text{energy}} &= C_{\text{loss}}, \\ C_{\text{delay}} &= C_{\text{base,delay}} + C_{\text{cong,delay}} + C_{\text{disp,delay}}, \\ C_{\text{soft}} &= C_{\text{operational}} + C_{\text{energy}} + C_{\text{delay}}. \end{aligned}$$

Component PV equations:  $C_{\text{O\&M}}$  (§B.2.1),  $C_{\text{insurance}}$  (§B.2.2),  $C_{\text{rent,total}}$  (§B.2.3),  $C_{\text{loss}}$  (§B.2.5),  $C_{\text{base,delay}}$  (§B.2.6),  $C_{\text{cong,delay}}$  (§B.2.7),  $C_{\text{disp,delay}}$  (§C.3.1).

## B.3 Risk Costs ( $C_{\text{risk}}$ )

These two sub-items produce the risk cost category  $C_{\text{risk}}$  (§B.3.2).

### B.3.1 Expected cost of wildfires

| Variable                  | Description                                                                                                                                                                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $L$                       | Total line length (miles). Sum of all terrain segment lengths from physical details.                                                                                                                                                             |
| $f$                       | Base ignition rate (events/(mi·yr)). Overhead baseline; line-level scalar.                                                                                                                                                                       |
| $k_t$                     | Construction-type multiplier (dimensionless). Overhead 1.0; underground 0.01 (94–99% reduction; Pacific Gas and Electric Company, Southern California Edison Company, and San Diego Gas & Electric Company 2025); subsea 0.0 (no ignition risk). |
| $f_t$                     | Effective ignition rate for type $t$ (events/(mi·yr)). $f_t = f \times k_t$ .                                                                                                                                                                    |
| $\lambda_{\text{wf}}$     | Total annual event rate (events/year). $\lambda_{\text{wf}} = L \times f_t$ .                                                                                                                                                                    |
| $S_{\text{event,wf}}$     | Severity (loss per event) (\$). Expected loss per wildfire event.                                                                                                                                                                                |
| $EAL_{\text{wf}}$         | Expected annual loss (\$/year). $EAL_{\text{wf}} = \lambda_{\text{wf}} \times S_{\text{event,wf}}$ .                                                                                                                                             |
| $g_{\text{wf}}$           | Wildfire risk growth rate (decimal). Annual increase in expected loss (e.g. escalation of risk).                                                                                                                                                 |
| $T_{\text{lifetime}}$     | Project lifetime (years).                                                                                                                                                                                                                        |
| $T_{\text{COD}}$          | Commercial operation date (year index) (years). First year of operation; cost stream starts here.                                                                                                                                                |
| $T_{\text{delay}}$        | Delay years. Used in PV discounting.                                                                                                                                                                                                             |
| $T_{\text{construction}}$ | Construction years. Used in PV discounting.                                                                                                                                                                                                      |
| $r_{\text{social}}$       | Social discount rate (decimal). Used to discount this externality cost (§A.2).                                                                                                                                                                   |
| $C_{\text{wf,nominal}}$   | Expected wildfire cost (nominal) (\$). Sum of growing annual $EAL_{\text{wf}}$ over lifetime; undiscounted.                                                                                                                                      |
| $C_{\text{wf}}$           | Expected wildfire cost (PV) (\$). PV of growing $EAL_{\text{wf}}$ stream from $T_{\text{COD}}$ over $T_{\text{lifetime}}$ ; discounted for delay and construction.                                                                               |

## Variables

## Equations

### 1. Effective ignition rate (construction type).

$$f_t = f \times k_t.$$

### 2. Total annual event rate.

$$\lambda_{\text{wf}} = L \times f_t.$$

### 3. Expected annual loss.

$$EAL_{\text{wf}} = \lambda_{\text{wf}} \times S_{\text{event, wf}}.$$

### 4. Nominal (undiscounted) total cost. Growing annual loss at rate $g_{\text{wf}}$ over $T_{\text{lifetime}}$ years:

$$C_{\text{wf, nominal}} = EAL_{\text{wf}} \times \frac{(1 + g_{\text{wf}})^{T_{\text{lifetime}}} - 1}{g_{\text{wf}}}.$$

**5. Real (present value) cost.** Two reference dates: Year 0 = project analysis start (permitting, pre-construction);  $T_{\text{COD}}$  = commercial operation date = first year the cost stream begins ( $T_{\text{delay}} + T_{\text{construction}}$  years after Year 0). Present value is expressed in Year-0 dollars.

The numerator is the present value of the growing annuity of annual losses ( $EAL_{\text{wf}}$  growing at  $g_{\text{wf}}$ ) over  $T_{\text{lifetime}}$  years, with “present” taken at  $T_{\text{COD}}$ . It is the lump sum at  $T_{\text{COD}}$  equivalent to the stream from  $T_{\text{COD}}$  to end of life. Value at  $T_{\text{COD}}$ :

$$\text{PV at } T_{\text{COD}} = EAL_{\text{wf}} \times \frac{1 - \left( \frac{1 + g_{\text{wf}}}{1 + r_{\text{social}}} \right)^{T_{\text{lifetime}}}}{r_{\text{social}} - g_{\text{wf}}}.$$

The denominator moves that lump sum from  $T_{\text{COD}}$  back to Year 0: the amount at  $T_{\text{COD}}$  is discounted by  $(1 + r_{\text{social}})^{T_{\text{delay}} + T_{\text{construction}}}$ .

The real (present value) cost is therefore:

$$C_{\text{wf}} = \frac{EAL_{\text{wf}} \times \frac{1 - \left( \frac{1 + g_{\text{wf}}}{1 + r_{\text{social}}} \right)^{T_{\text{lifetime}}}}{r_{\text{social}} - g_{\text{wf}}}}{(1 + r_{\text{social}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

### Default calibration

The default base ignition rate  $f = 0.002$  events/(mi-yr) is derived from CPUC-reportable transmission ignition data: PG&E HFTD ignition data (Pacific Gas and Electric Company 2025), while SDG&E CPUC filing data yield 0.00244 fires/mi/yr for all-cause transmission ignitions (Johnson 2014). Cross-check: applying the 3:1 distribution-to-transmission ratio (Taylor and Roald 2022) to distribution rates (Warner, Callaway, and Fowle 2025) yields  $\sim 0.0017$ /mi/yr, consistent with the chosen default. The default severity  $S_{\text{event, wf}} = \$8,328,765$  per event (2025\$) is a probability-weighted average across fire size classes, anchored by the finding that fewer than 1% of grid-caused ignitions exceed 10,000 acres yet account for 79% of total acreage burned (Eagleton et al. 2025). Three independent cross-checks — PG&E total liability per ignition (\$6–7M), California annual property damage per ignition (\$4.5M), and a bottom-up size-class decomposition (\$13M) — bracket the recommended value. The risk growth rate  $g_{\text{wf}} = 1.5\%$ /yr reflects a projected 45% cumulative increase in utility-ignited wildfire risk by 2050 (Warner, Callaway, and Fowle 2025). Defaults are calibrated to fire-prone overhead transmission in the western United States; parameters may be varied in sensitivity analysis for other regions or risk profiles.

### B.3.2 Expected cost of outages

| Variable                    | Description                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $CT$                        | Construction type index. e.g. overhead, underground, subsea.                                                                                                                                                                                     |
| $L$                         | Total line length (miles). Sum of all terrain segment lengths from physical details.                                                                                                                                                             |
| $r_{CT}$                    | Outage rate for construction type $CT$ (events/(mi-yr)). Underground/subsea: flat per-mile rate.                                                                                                                                                 |
| $\lambda_{\text{terminal}}$ | Terminal outage rate (outages/yr). Overhead only. Default 0.2136. Length-independent component (BC Hydro 2006).                                                                                                                                  |
| $\lambda_{\text{per mile}}$ | Per-mile outage rate (outages/(mi-yr)). Overhead only. Default 0.012. Exposure-dependent (Pijarski and Belowski 2025).                                                                                                                           |
| $\lambda_{\text{out}}$      | Total annual outage rate (events/year). Overhead: $\lambda_{\text{out}} = \lambda_{\text{terminal}} + \lambda_{\text{per mile}} \times L$ . Underground/subsea: $\lambda_{\text{out}} = L \times r_{CT}$ .                                       |
| $H_{\text{base}}$           | Base outage duration (h/event). Hours per outage event; line-level scalar.                                                                                                                                                                       |
| $k_{CT}$                    | Duration multiplier for construction type $CT$ (dimensionless). Overhead 1; underground, subsea can be $> 1$ .                                                                                                                                   |
| $H_{\text{eff}}$            | Effective outage duration (h/event). $H_{\text{eff}} = H_{\text{base}} \times k_{CT}$ .                                                                                                                                                          |
| $\phi$                      | Capacity-at-risk factor (dimensionless). $\phi = 1/N_{\text{poles}}$ , where $N_{\text{poles}}$ is the number of circuits (AC) or poles (DC) from the conductor specification. For single-circuit AC, $\phi = 1$ ; for bipole DC, $\phi = 0.5$ . |
| $C$                         | Line capacity (MW). Project capacity (nameplate).                                                                                                                                                                                                |
| $u$                         | Line utilization factor (dimensionless). Average fraction of line capacity in use; from project technical details.                                                                                                                               |
| $MW_{\text{lost}}$          | MW lost per event (MW). $MW_{\text{lost}} = \phi \times C \times u$ .                                                                                                                                                                            |
| $\rho$                      | Load-shed fraction (dimensionless). Fraction of lost flow that becomes unserved energy. AC meshed: $\rho = 0.05$ ; DC point-to-point: $\rho = 0.80$ . Derived from AC/DC classification.                                                         |
| $c_{\text{redisp}}$         | Redispatch cost (\$/MWh). Marginal congestion cost of rerouting power through alternative paths. Default \$24/MWh (2025\$).                                                                                                                      |
| $U$                         | Unserved energy per event (MWh/event). $U = H_{\text{eff}} \times \rho \times MW_{\text{lost}}$ .                                                                                                                                                |
| $v_k$                       | VoLL for duration tier $k$ (\$/MWh). 10 tiers from 0–1 h through 720+ h; see VoLL tier table below.                                                                                                                                              |
| $C_{\text{event,out}}$      | Cost per outage event, \$/event. Sum of load-shed cost (piecewise VoLL on $\rho \cdot MW_{\text{lost}}$ ) and redispatch cost ( $c_{\text{redisp}}$ on $(1 - \rho) \cdot MW_{\text{lost}}$ ).                                                    |
| $EAL_{\text{out}}$          | Expected annual loss (\$/year). $EAL_{\text{out}} = \lambda_{\text{out}} \times C_{\text{event,out}}$ .                                                                                                                                          |
| $g_{\text{outages}}$        | Outage risk growth rate (decimal). Annual increase in expected cost (optional).                                                                                                                                                                  |
| $T_{\text{lifetime}}$       | Project lifetime (years).                                                                                                                                                                                                                        |
| $T_{\text{COD}}$            | Commercial operation date (year index) (years). First year of operation; cost stream starts here.                                                                                                                                                |
| $T_{\text{delay}}$          | Delay years. Used in PV discounting.                                                                                                                                                                                                             |
| $T_{\text{construction}}$   | Construction years. Used in PV discounting.                                                                                                                                                                                                      |
| $r_{\text{social}}$         | Social discount rate (decimal). Used to discount this externality cost (§A.2).                                                                                                                                                                   |
| $C_{\text{outage,nominal}}$ | Expected outage cost (nominal) (\$). Sum of growing annual $EAL_{\text{out}}$ over lifetime; undiscounted.                                                                                                                                       |
| $C_{\text{outage}}$         | Expected outage cost (PV) (\$). PV of growing $EAL_{\text{out}}$ stream from $T_{\text{COD}}$ over $T_{\text{lifetime}}$ ; discounted for delay and construction.                                                                                |

#### Variables

#### Equations

**1. Outage rate (by construction type).** For overhead lines, the outage rate follows a decomposition model capturing length-independent terminal effects and exposure-dependent line failures:

$$\lambda_{\text{out}} = \lambda_{\text{terminal}} + \lambda_{\text{per mile}} \times L \quad (\text{overhead}).$$

For underground and subsea, the pure per-mile linear model applies:

$$\lambda_{\text{out}} = L \times r_{CT} \quad (\text{underground, subsea}).$$

Effective outage duration:  $H_{\text{eff}} = H_{\text{base}} \times k_{CT}$ .

The decomposition separates length-independent terminal failures ( $\lambda_{\text{terminal}}$ ) from exposure-dependent per-mile failures ( $\lambda_{\text{per mile}}$ ); sources and calibration values are in the defaults block below.

**2. MW lost and unserved energy per event.**

$$MW_{\text{lost}} = \phi \times C \times u, \quad U = H_{\text{eff}} \times \rho \times MW_{\text{lost}}.$$

**3. Cost per event (piecewise VoLL).** Duration is split into tiers with VoLL (\$/MWh) per tier. Tiers 1–5 are ERCOT system-wide survey values (Gibbons and Sergici 2024); tiers 6–9 are extrapolated; tier 10 is a flat cap at the 30-day value. Values are 2025\$, escalated from (Gibbons and Sergici 2024).

| Tier | Duration | $v_k$ (\$/MWh) | Source       |
|------|----------|----------------|--------------|
| 1    | 0–1 h    | 36,620         | Observed     |
| 2    | 1–2 h    | 21,885         | Observed     |
| 3    | 2–4 h    | 13,689         | Observed     |
| 4    | 4–8 h    | 10,708         | Observed     |
| 5    | 8–16 h   | 13,937         | Observed     |
| 6    | 16–32 h  | 20,628         | Extrapolated |
| 7    | 32–64 h  | 36,282         | Extrapolated |
| 8    | 64 h–7 d | 88,512         | Extrapolated |
| 9    | 7–30 d   | 367,280        | Extrapolated |
| 10   | 30 d+    | 367,280        | Cap          |

**VoLL extrapolation model.** Observed ERCOT system-wide VoLL (\$/MWh) declines sharply from 1 to 8 hours, then reverses at 16 hours. To extend beyond the 16-hour survey ceiling, a three-term model is fitted by ordinary least squares to the five observed values:

$$V(t) = \frac{a}{t} + b + ct, \quad a = 32,210, \quad b = 3,339, \quad c = 492.$$

The  $a/t$  term captures front-loaded disruption costs (production shutdown, restart);  $b$  is a steady-state floor;  $ct$  captures escalating societal costs for extended outages (food safety, medical, economic cascading). The model minimum occurs at  $t^* = \sqrt{a/c} \approx 8.1$  hours, consistent with the observed trough at 8 hours. For  $t > 720$  hours (30 days), VoLL is capped at \$367,280/MWh; outages beyond 30 days are assumed to transition from outage recovery to grid reconstruction, a regime outside the scope of this model.

Cost per event is the sum of a load-shed term (piecewise VoLL applied to the shed fraction) and a redispatch term (flat congestion cost applied to the rerouted fraction):

$$C_{\text{event,out}} = \underbrace{\sum_k (\text{hours in tier } k) \times \rho \cdot MW_{\text{lost}} \times v_k}_{\text{load-shed cost}} + \underbrace{(1 - \rho) \cdot MW_{\text{lost}} \cdot H_{\text{eff}} \cdot c_{\text{redispatch}}}_{\text{redispatch cost}}.$$

**Two-tier outage cost decomposition.** In a well-planned meshed network, N-1 contingency criteria (North American Electric Reliability Corporation (NERC) TPL-001-5.1) require that no single-element outage causes involuntary load shedding under normal conditions. Systems meeting the standard loss-of-load expectation of 0.1 days/year exhibit normalized expected unserved energy of only 1–10 parts per million of total demand. The economic consequence of most transmission outages is therefore not load shedding but congestion: the cost of redispatching more expensive generation to compensate for the lost transfer capability.

The outage cost per event is accordingly decomposed into two terms. The first term values the fraction  $\rho$  of lost flow that becomes actual unserved energy at the piecewise VoLL schedule. The second term values the remainder  $(1 - \rho)$  at the marginal redispatch cost  $c_{\text{redispatch}}$ , representing the congestion premium during the outage. This two-tier structure follows the ENTSO-E cost-benefit analysis framework (ENTSO-E 2021), which separately quantifies socio-economic welfare from congestion relief (indicator B1) and security-of-supply benefit from avoided load shedding (indicator B6), summing them without double-counting.

The load-shed fraction  $\rho$  is derived from AC/DC classification:  $\rho = 0.05$  for AC lines in meshed networks, reflecting N-1 compliance and expected-unserved-energy standards; and  $\rho = 0.80$  for DC point-to-point lines, reflecting limited alternative paths with partial reserve margin at the receiving end. The default redispatch cost  $c_{\text{redispatch}} = \$24/\text{MWh}$  (2025\$) is the interregional median of empirical congestion relief values from LBNL analysis of historical locational marginal prices across U.S. ISOs, which finds median values of \$11/MWh for regional links and \$24/MWh for interregional links (Millstein et al. 2022). U.S. system-wide transmission congestion costs of \$6–21 billion annually (Shreve et al. 2025), across approximately 4,000 TWh of consumption, corroborate a system-average congestion cost of \$1.5–5/MWh.

#### 4. Expected annual loss.

$$EAL_{\text{out}} = \lambda_{\text{out}} \times C_{\text{event,out}}.$$

#### 5. Nominal (undiscounted) total cost.

Growing annual cost at rate  $g_{\text{outages}}$  over  $T_{\text{lifetime}}$  years:

$$C_{\text{outage,nominal}} = EAL_{\text{out}} \times \frac{(1 + g_{\text{outages}})^{T_{\text{lifetime}}} - 1}{g_{\text{outages}}}.$$

**6. Real (present value) cost.** Same growing-annuity PV as in §B.3.1 (paragraph 5), with  $EAL_{\text{out}}$  replacing  $EAL_{\text{wf}}$ ,  $g_{\text{outages}}$  replacing  $g_{\text{wf}}$ , and  $r_{\text{social}}$  as the discount rate:

$$C_{\text{outage}} = \frac{EAL_{\text{out}} \times \frac{1 - \left( \frac{1 + g_{\text{outages}}}{1 + r_{\text{social}}} \right)^{T_{\text{lifetime}}}}{r_{\text{social}} - g_{\text{outages}}}}{(1 + r_{\text{social}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

**Note on VoLL.** The piecewise VoLL schedule applies to the load-shed fraction  $\rho \cdot MW_{\text{lost}}$ ; the redispatch fraction  $(1 - \rho) \cdot MW_{\text{lost}}$  is valued at the flat rate  $c_{\text{redisp}}$ . Capacity at risk  $\phi$  (e.g. 1 for radial,  $< 1$  for meshed/redundant) is the fraction of line capacity assumed lost per outage event.

#### Default calibration

The default overhead outage model uses a decomposition  $\lambda = 0.2136 + 0.012 \times L$ , giving an effective rate of  $\sim 0.014/\text{mi}$  for a 100-mile line,  $\sim 0.023/\text{mi}$  for a 20-mile line, and  $\sim 0.013/\text{mi}$  for a 300-mile line. The terminal component (0.2136 outages/yr) captures length-independent substation and protection equipment failures (BC Hydro 2006), consistent with NERC TADS data showing  $\sim 23\%$  of sustained AC-circuit outages originate from substation and circuit equipment failures (NERC 2025); the per-mile component (0.012/mi/yr) reflects exposure-dependent conductor and insulator failures, informed by Polish transmission system operator (TSO) multi-year averages (220 kV: 0.012/mi/yr; Pijarski and Belowski 2025). Aggregating NERC TADS event rates across initiating-cause codes (NERC 2025) implies a national transmission outage rate of  $\sim 0.010/\text{mi/yr}$ . The underground repair duration multiplier  $k_{\text{underground}} = 80$  (effective duration  $6 \text{ h} \times 80 = 480 \text{ h} \approx 20 \text{ days}$ ) reflects ENTSO-E/Europacable (2019) average XLPE land-cable repair times, consistent with Ofgem/GHD (2020) modeled repair durations (20 days XLPE, 40 days mass-impregnated) and Wisconsin PSC guidance characterizing underground repairs as spanning weeks to months. The subsea outage rate default  $r_{\text{subsea}} = 0.0007 \text{ outages}/(\text{mi} \cdot \text{yr})$  reflects Ofgem/GHD (2020) risk-assessed total failure rate data (0.000615 failures/(yr·cct-km), combined internal and external faults) and ENTSO-E/Europacable (2019) HVDC cable-only reliability data (0.07 faults/100 km/yr), which cites CIGRÉ TB 379 (2009) baseline survey data showing reduced submarine cable failure rates relative to earlier estimates. The subsea duration multiplier  $k_{\text{subsea}} = 250$  (effective duration  $\sim 62 \text{ days}$ ) reflects the 60-day average repair duration per ENTSO-E/Europacable (2019) (ENTSO-E and Europacable 2019); longer repair episodes (3–5 months) documented for offshore wind export cables in IEA Wind TEM95 proceedings (McKeever and Wyatt 2019) reflect weather and logistics delays beyond the baseline average. The underground land cable rate ( $\lambda_{\text{UG}} = 0.004/\text{mi/yr}$ ) draws on XLPE land cable failure rates of 0.00069/cct-km/yr (internal, (ENTSO-E and Europacable 2019)) to 0.00088/cct-km/yr (all-cause, (CIGRE Working Group B1.10 2009)). FORGE's value (0.0025/km) is approximately 2–3 $\times$  the European cable-body baseline, reflecting a U.S. system-level rate that includes joint, termination, and third-party-damage failures—consistent with NERC TADS underground circuit outage data of  $\sim 0.003\text{--}0.005/\text{mi/yr}$  (2013–2018 average). Value of lost load tiers are directly calibrated to the ERCOT system-wide customer survey (Gibbons and Sergici 2024), which reports VOLL by duration from 1 to 16 hours (e.g. \$36,620/MWh at 1 hour, \$10,708/MWh at 8 hours, 2025\$). Beyond the 16-hour survey ceiling, VoLL is extrapolated using the three-term model  $V(t) = 32,210/t + 3,339 + 492t$ , capped at 30 days. The LBNL ICE Calculator (Sullivan, Schellenberg, and Blundell 2015; Larsen et al. 2025) provides corroborating estimates by customer class. The load-shed fraction  $\rho$  is determined by AC/DC classification ( $\rho = 0.05$  for AC,  $\rho = 0.80$  for DC). The redispatch cost  $c_{\text{redisp}}$  defaults to \$24/MWh (Millstein et al. 2022); the parameter may be varied for projects with known congestion exposure.

#### Risk cost total

The two risk-cost components above sum to the risk cost category. Risk costs are probabilistic and event-driven.

$$C_{\text{risk}} = C_{\text{wf}} + C_{\text{outage}}.$$

Component PV equations:  $C_{\text{wf}}$  (§B.3.1),  $C_{\text{outage}}$  (§B.3.2).

**Note on independence of wildfire and outage costs.** A single wildfire event can simultaneously cause property and environmental damage (counted in  $C_{\text{wf}}$ ) and a transmission outage (counted in  $C_{\text{outage}}$ ), either directly through line damage or indirectly through preventive de-energization (public safety power shutoff, PSPS). Despite this shared triggering event, independent treatment is defensible because the two cost categories capture distinct consequence dimensions:  $C_{\text{wf}}$  values property destruction, suppression expenditures, and environmental losses, whereas  $C_{\text{outage}}$  values the economic cost of unserved energy to consumers via VoLL. Additive treatment may overstate total risk cost for projects in high-wildfire zones where fire-caused outages constitute a significant fraction of all outage events.

#### B.4 Emissions Costs ( $C_{\text{emissions}}$ )

$C_{\text{emissions}}^P = C_{\text{comp,emissions}}^P$  (loss-compensation emissions only). The facilitated emissions costs  $C_{\text{fac,emissions}}^P$  and  $C_{\text{fac,emissions}}^{\text{no}}$  are intermediate quantities whose difference forms the avoided-emissions benefit  $B_{\text{avoided,emissions}}^P$  (§C.3). Both calculations use the same rate-based mix-evolution equations, emission intensities  $I_{j,k}$ , societal costs  $c_k(\tau)$ , and social discount rate  $r_{\text{social}}$ . Note: emissions reductions are not among the seven required benefit categories in FERC Order No. 1920 (May 2024) but are permitted as optional additional benefits consistent with Order Nos. 890 and 1000.

### B.4.1 Loss-compensation emissions

The societal cost of emissions ( $\text{CO}_2$ ,  $\text{SO}_x$ ,  $\text{NO}_x$ ) from extra generation used to compensate for transmission energy losses. Energy losses ( $E_{\text{loss}}$ ) are defined in §B.2.4; this subsection only adds externality valuation through the grid’s time-varying fuel mix and pollutant prices. Under the single-trajectory grid model (defined in §B.4.2), shares  $p_j(\tau)$  here are the with-line grid shares  $p_j^{\text{with}}(\tau)$ : the regional grid mix evolving under the line’s influence during operations, starting from the shared grid state at COD. The without-line path  $p_j^{\text{no}}(\tau)$  used by the facilitated-emissions and displacement subsections does not enter loss-compensation calculations.

| Variable                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_{\text{loss}}$                   | Total energy losses (annual) (MWh/year). From §B.2.4.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| $\alpha$                            | Compensation share (dimensionless [0,1]). Fraction of $E_{\text{loss}}$ compensated by additional generation.                                                                                                                                                                                                                                                                                                                                                                                   |
| $TEC$                               | Total energy compensated (annual) (MWh/year). $TEC = \alpha \times E_{\text{loss}}$ .                                                                                                                                                                                                                                                                                                                                                                                                           |
| $j$                                 | Energy source index (index). e.g. coal, gas, solar, wind, hydro, nuclear.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $p_j^{\text{with}}(\tau)$           | With-line grid share of source $j$ in operating year $\tau$ (dimensionless). Grid state at COD, $p_j(T_{\text{COD}})$ (§B.4.2), evolved at the post-COD per-source mix-evolution rates $g_j^{\text{post}}$ ; $\sum_j p_j^{\text{with}}(\tau) = 1$ . See §B.4.2 for the without-line path and the COD handoff.                                                                                                                                                                                   |
| $I_{j,k}$                           | Emission intensity (kg/MWh). Emissions of pollutant $k$ per MWh from source $j$ . For $\text{CO}_2$ : lifecycle medians from Bruckner et al. (2014), Table A.III.2. For $\text{SO}_x/\text{NO}_x$ : eGRID subregion average combustion emission rates (EPA 2025b), applied uniformly across generation types rather than as fuel-specific lifecycle values.                                                                                                                                     |
| $k$                                 | Pollutant index (index). $\text{CO}_2$ , $\text{SO}_x$ , $\text{NO}_x$ .                                                                                                                                                                                                                                                                                                                                                                                                                        |
| $E_{k,\tau}$                        | Emissions of pollutant $k$ in year $\tau$ (kg). From compensated energy and mix.                                                                                                                                                                                                                                                                                                                                                                                                                |
| $c_k$                               | Societal cost per kg of pollutant $k$ (\$/kg). Year-1 level, 2025\$. Default: \$236/t $\text{CO}_2$ (Rennert’s \$185/t in 2020\$ (Rennert et al. 2022), escalated to 2025\$ via SC- $\text{CO}_2$ growth schedule; cf. (EPA 2023)); $\text{SO}_x$ \$76/kg, $\text{NO}_x$ \$143/kg (EPA BPT (U.S. Environmental Protection Agency 2023), EGU sector, analysis year 2025, 3% discount rate; $\text{NO}_x$ sums $\text{PM}_{2.5}$ -related and ozone-related columns; 2019\$ escalated to 2025\$). |
| $g_{c,k}$                           | Real annual escalation rate for $c_k$ (decimal). Default 0.02 for $\text{CO}_2$ , 0 for $\text{SO}_x/\text{NO}_x$ .                                                                                                                                                                                                                                                                                                                                                                             |
| $c_k(\tau)$                         | Societal cost per kg in operating year $\tau$ (\$/kg). $c_k(1) \cdot (1 + g_{c,k})^{\tau-1}$ .                                                                                                                                                                                                                                                                                                                                                                                                  |
| $C_{k,\tau}$                        | Cost of pollutant $k$ in year $\tau$ (\$). $E_{k,\tau} \times c_k(\tau)$ .                                                                                                                                                                                                                                                                                                                                                                                                                      |
| $C_\tau$                            | Total emissions cost in year $\tau$ (\$/year). Sum over pollutants.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| $T_{\text{lifetime}}$               | Project lifetime (years).                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $T_{\text{COD}}$                    | Commercial operation date (year). First year of operation.                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| $r_{\text{social}}$                 | Social discount rate (decimal). Used to discount this externality (§A.2).                                                                                                                                                                                                                                                                                                                                                                                                                       |
| $C_{\text{comp,emissions,nominal}}$ | Nominal lifetime loss-compensation emissions cost (\$). Undiscounted sum over lifetime.                                                                                                                                                                                                                                                                                                                                                                                                         |
| $C_{\text{comp,emissions}}$         | Loss-compensation emissions cost (PV) (\$). PV of annual stream from $T_{\text{COD}}$ over $T_{\text{lifetime}}$ .                                                                                                                                                                                                                                                                                                                                                                              |

**Variables** Default  $\text{CO}_2$  intensities  $I_{j,\text{CO}_2}$  (kg/MWh) by fuel type are lifecycle medians from IPCC AR5 (Bruckner et al. 2014).  $\text{SO}_x$  and  $\text{NO}_x$  intensities are fuel-specific combustion emission factors consistent with EPA eGRID reporting (EPA 2025b); renewables, nuclear, and hydro are zero by definition:

Table 4: Default emission intensities  $I_{j,k}$  (kg/MWh) by fuel type.

| Fuel        | $\text{CO}_2$ | $\text{SO}_x$ | $\text{NO}_x$ |
|-------------|---------------|---------------|---------------|
| Coal        | 820           | 13.0          | 4.0           |
| Oil         | 650           | 10.0          | 3.5           |
| Natural gas | 490           | 0.1           | 0.2           |
| Solar       | 48            | 0             | 0             |
| Wind        | 11            | 0             | 0             |
| Hydro       | 24            | 0             | 0             |
| Nuclear     | 12            | 0             | 0             |

### Equations

#### 1. Total energy compensated (annual).

$$TEC = \alpha \times E_{\text{loss}}.$$

**2. Energy mix by operating year.** For each operating year  $\tau = 1, \dots, T_{\text{lifetime}}$ , with-line grid shares  $p_j^{\text{with}}(\tau)$  evolve from the grid state at COD,  $p_j(T_{\text{COD}})$  (§B.4.2, Eq. 2), at the post-COD rates  $g_j^{\text{post}}$ , normalized so

$$\sum_j p_j^{\text{with}}(\tau) = 1.$$

### 3. Emissions in operating year $\tau$ (by pollutant).

$$E_{k,\tau} = TEC \times \sum_j p_j^{\text{with}}(\tau) \times I_{j,k}.$$

### 4. Societal cost schedule.

$$c_k(\tau) = c_k(1) \cdot (1 + g_{c,k})^{\tau-1}.$$

### 5. Cost in operating year $\tau$ (by pollutant).

$$C_{k,\tau} = E_{k,\tau} \times c_k(\tau).$$

### 6. Total emissions cost in operating year $\tau$ .

$$C_\tau = \sum_k C_{k,\tau}.$$

### 7. Nominal (undiscounted) lifetime cost.

$$C_{\text{comp,emissions,nominal}} = \sum_{\tau=1}^{T_{\text{lifetime}}} C_\tau.$$

### 8. Present value.

$$C_{\text{comp,emissions}} = \sum_{\tau=1}^{T_{\text{lifetime}}} \frac{C_\tau}{(1 + r_{\text{social}})^{T_{\text{COD}} + \tau - 1}}.$$

**Time-varying societal cost and discounting.** The societal cost  $c_k(\tau)$  escalates in real terms following standard integrated-assessment-model damage pathways (cf. (Rennert et al. 2022)). The escalation rate  $g_{c,k}$  is distinct from the social discount rate  $r_{\text{social}}$ :  $g_{c,k}$  reflects the physical trajectory of damage costs, while  $r_{\text{social}}$  discounts the annual cost stream to the project’s present value. These operate on different objects and do not constitute double-discounting.

Both  $C_{\text{energy}}$  and  $C_{\text{comp,emissions}}$  originate from  $E_{\text{loss}}$  (physical transmission losses), but they use different valuations:  $C_{\text{energy}}$  applies the value of load to the MWh of lost energy;  $C_{\text{comp,emissions}}$  applies externality prices to the pollutant emissions from the make-up generation. They measure different economic effects and do not double-count.

**Grid mix assumption for loss compensation.** Loss-compensation emissions are computed using the post-COD regional grid mix, reflecting marginal system dispatch. For radial lines delivering a specific resource (e.g., a dedicated wind corridor), this assumes losses are compensated by the regional fleet rather than by the resource itself. This is conservative (the regional fleet is more carbon-intensive than the dedicated resource) and consistent with the system-dispatch perspective: the generating resource produces independently of line losses, and the system compensates the delivery shortfall at the margin.

## B.4.2 Facilitated emissions

The absolute facilitated emissions costs  $C_{\text{fac,emissions}}^P$  and  $C_{\text{fac,emissions}}^{\text{no}}$  are intermediate quantities computed here; they do not enter the cost category directly. Their difference forms the avoided-emissions benefit  $B_{\text{avoided,emissions}}^P$  (§C.3).

**Single-trajectory grid model.** The CTCC represents the regional grid’s fuel mix as *one* shared trajectory rather than two independent hypothetical grids. There is a single set of initial generation shares  $p_j(0)$  at year 0. Two exogenous growth-rate regimes describe how that one grid can evolve: the *pre-COD* rates  $g_j^{\text{pre}}$  (how the grid changes on its own, absent the line) and the *post-COD* rates  $g_j^{\text{post}}$  (how the line’s influence bends that evolution once it is operating). Both the with-line and without-line paths are computed from the *same* starting shares and *share* an identical state at commercial operation (COD) — the grid is whatever it organically became during permitting and construction delay, since the line has not yet influenced it. The two paths diverge only afterward, one evolving at  $g_j^{\text{pre}}$  and the other at  $g_j^{\text{post}}$ . This single-trajectory design—as opposed to two fully independent mix series with different starting points—eliminates an artificial discontinuity at COD and ensures the counterfactual grid is causally consistent with the project’s own delay.

The absolute social cost of emissions from each trajectory’s grid path over the deliverable energy  $E_{\text{delivered,annual}}$  (§C.2.1) follows from this model:  $C_{\text{fac,emissions}}^P$  (with-line) evolves the COD state forward at  $g_j^{\text{post}}$ ;  $C_{\text{fac,emissions}}^{\text{no}}$  (without-line) evolves it forward at  $g_j^{\text{pre}}$ . Both enter the avoided-emissions calculation (§C.3).

**Non-double-counting with loss compensation.** The loss-compensation subsection (§B.4.1) and this subsection operate on disjoint MWh pools:  $E_{\text{loss}}$  (energy dissipated along the line) vs.  $E_{\text{delivered,annual}}$  (throughput reaching the load end). Setting  $\alpha = 0$  removes the loss-compensation stream without affecting facilitated emissions, and vice versa. No MWh appears in both formulas.

**Non-double-counting with delivered-energy market benefit.** The same  $E_{\text{delivered,annual}}$  carries one market valuation ( $v_{\text{load}}$  in §C.2.1) and one externality valuation ( $c_k$  per kg here). These measure different economic objects: willingness-to-pay for delivered energy vs. social cost of pollutant emissions.

**Non-double-counting with remedial benefits.** Curtailment is not modeled as a separate remedial category, so the former overlap between curtailment benefit and delivered energy (both pricing energy value of constrained-hour MWh) is removed by construction. Remedial benefit is congestion relief only: constrained-hour MWh valued at the redispatch premium  $\gamma_{\text{congestion}}$ , which is a different economic object from the energy's own value  $v_{\text{load}}$ .

**Transparency: value-of-load internalization.** If  $v_{\text{load}}$  already partially reflects emissions costs (e.g. through carbon pricing, cap-and-trade allowance costs, or renewable energy credits), part of the externality may already be captured in the delivered-energy benefit. To avoid double-counting, either (a) define  $v_{\text{load}}$  net of carbon/emissions adders, or (b) define  $c_k$  to exclude externalities already reflected in the value of load. The default assumption is that  $v_{\text{load}}$  reflects a wholesale-level willingness to pay that does not fully internalize the societal externalities valued by  $c_k$ .

| Variable                         | Description                                                                                                                                                                                |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_{\text{delivered,annual}}$    | Deliverable energy (MWh/year). From §C.2.1. Constant over lifetime.                                                                                                                        |
| $j$                              | Energy source index (index). Same as §B.4.1.                                                                                                                                               |
| $k$                              | Pollutant index (index). $\text{CO}_2$ , $\text{SO}_x$ , $\text{NO}_x$ ; same as §B.4.1.                                                                                                   |
| $p_j(0)$                         | Initial grid share of source $j$ (dimensionless). Exogenous initial regional generation shares; $\sum_j p_j(0) = 1$ . Single regional grid, shared by both trajectories.                   |
| $g_j^{\text{pre}}$               | Per-source annual rate of change without the line (exogenous). Governs the without-line trajectory continuously from year 0.                                                               |
| $g_j^{\text{post}}$              | Per-source annual rate of change with the line (exogenous). Governs the with-line trajectory from COD onward (and hypothetically during delay; §C.3.1).                                    |
| $p_j(T_{\text{COD}})$            | Grid share of source $j$ at COD (dimensionless). $= p_j^{\text{no}}(T_{\text{COD}})$ : the without-line trajectory evaluated at COD. Shared starting point for both operational instances. |
| $p_j^{\text{with}}(\tau)$        | With-line grid share of source $j$ , operating year $\tau$ . $p_j(T_{\text{COD}})$ evolved at $g_j^{\text{post}}$ ; same path used by loss-comp (§B.4.1).                                  |
| $p_j^{\text{no}}(\tau)$          | Without-line grid share of source $j$ , operating year $\tau$ . $p_j(T_{\text{COD}})$ evolved at $g_j^{\text{pre}}$ ; used here and by displacement (§C.3).                                |
| $I_{j,k}$                        | Emission intensity (kg/MWh). Shared with loss-compensation emissions (§B.4.1); lifecycle medians as cited there (e.g. Bruckner et al.).                                                    |
| $EF_k^{(s)}(\tau)$               | Implied emission factor (kg/MWh). For instance $s \in \{\text{with, no}\}$ .                                                                                                               |
| $E_{k,\tau}^{(s)}$               | Emissions mass under instance $s$ , year $\tau$ (kg).                                                                                                                                      |
| $C_{k,\tau}^{(s)}$               | Emissions cost under instance $s$ , year $\tau$ (\$).                                                                                                                                      |
| $PV_k^{(s)}$                     | PV emissions cost for pollutant $k$ , instance $s$ (\$).                                                                                                                                   |
| $c_k$                            | Societal cost per kg of pollutant $k$ (\$/kg). Year-1 level; same as §B.4.1.                                                                                                               |
| $g_{c,k}$                        | Real annual escalation rate for $c_k$ (decimal). Same as §B.4.1. Default 0.02 for $\text{CO}_2$ , 0 for $\text{SO}_x/\text{NO}_x$ .                                                        |
| $c_k(\tau)$                      | Societal cost per kg in operating year $\tau$ (\$/kg). $c_k(1) \cdot (1 + g_{c,k})^{\tau-1}$ .                                                                                             |
| $r_{\text{social}}$              | Social discount rate (decimal). Same as §B.4.1.                                                                                                                                            |
| $T_{\text{COD}}$                 | Commercial operation date (year). First year of operation.                                                                                                                                 |
| $T_{\text{lifetime}}$            | Project lifetime (years).                                                                                                                                                                  |
| $C_{\text{fac,emissions}}^{(s)}$ | Absolute facilitated emissions cost, instance $s$ (\$). $= \sum_k PV_k^{(s)}$ . Intermediate; does not enter $C_{\text{emissions}}^P$ directly.                                            |

## Variables

## Equations

**1. Scope (deliverable energy).** The scoped MWh is the deliverable energy from §C.2.1, constant over the project lifetime:

$$H_{S,\tau} = E_{\text{delivered,annual}}.$$

**2. Grid state at COD (single-trajectory handoff).** Absent the line, the grid evolves continuously at  $g_j^{\text{pre}}$  from the initial shares  $p_j(0)$ . This same without-line path also defines the grid's actual state the moment the line reaches commercial operation. Compound growth/decay from year 0 to  $T_{\text{COD}}$ :

$$\tilde{p}_j(T_{\text{COD}}) = p_j(0) \cdot (1 + g_j^{\text{pre}})^{T_{\text{COD}}-1}.$$

Renormalize:

$$p_j(T_{\text{COD}}) = \frac{\tilde{p}_j(T_{\text{COD}})}{\sum_{j'} \tilde{p}_{j'}(T_{\text{COD}})}.$$

$p_j(T_{\text{COD}})$  is the shared starting point for both operational trajectories below.

**3. Operational mix evolution (both instances, from the COD state).** For each trajectory  $s \in \{\text{with}, \text{no}\}$ , starting from the common COD state  $p_j(T_{\text{COD}})$ , the same rate-based compound-growth equations used in §B.4.1 yield operating-year time paths. Compound growth/decay:

$$\tilde{p}_j^{(s)}(\tau) = p_j(T_{\text{COD}}) \cdot (1 + g_j^{(s)})^{\tau-1}, \quad \tau = 1, \dots, T_{\text{lifetime}}.$$

Renormalize each year:

$$p_j^{(s)}(\tau) = \frac{\tilde{p}_j^{(s)}(\tau)}{\sum_{j'} \tilde{p}_{j'}^{(s)}(\tau)}.$$

The with-line trajectory uses  $g_j^{(\text{with})} = g_j^{\text{post}}$ ; the without-line trajectory uses  $g_j^{(\text{no})} = g_j^{\text{pre}}$ .

**4. Implied emission factor.** For  $s \in \{\text{with}, \text{no}\}$  and pollutant  $k$ :

$$EF_k^{(s)}(\tau) = \sum_j p_j^{(s)}(\tau) I_{j,k}.$$

**5. Absolute emissions mass per year (by instance).** For instance  $s$  and pollutant  $k$ :

$$E_{k,\tau}^{(s)} = E_{\text{delivered,annual}} \times EF_k^{(s)}(\tau).$$

**6. Societal cost schedule.**

$$c_k(\tau) = c_k(1) \cdot (1 + g_{c,k})^{\tau-1}.$$

**7. Absolute emissions cost per year.**

$$C_{k,\tau}^{(s)} = E_{k,\tau}^{(s)} \times c_k(\tau).$$

**8. Present value per pollutant (by instance).** Discounted at  $r_{\text{social}}$  with the same index convention as §B.4.1:

$$PV_k^{(s)} = \sum_{\tau=1}^{T_{\text{lifetime}}} \frac{C_{k,\tau}^{(s)}}{(1 + r_{\text{social}})^{T_{\text{COD}} + \tau - 1}}.$$

**9. Total across pollutants — absolute facilitated emissions cost.**

$$C_{\text{fac,emissions}}^{(s)} = \sum_k PV_k^{(s)}.$$

Pollutants:  $\text{CO}_2$ ,  $\text{SO}_x$ ,  $\text{NO}_x$ —the same set as §B.4.1.  $C_{\text{fac,emissions}}^P$  (with-line,  $s = \text{with}$ ) and  $C_{\text{fac,emissions}}^{\text{no}}$  are intermediate quantities; their difference forms  $B_{\text{avoided,emissions}}^P$  (§C.3).

**Emissions cost aggregate**

$$C_{\text{emissions}}^P = C_{\text{comp,emissions}}^P.$$

See §B.4.1 for  $C_{\text{comp,emissions}}^P$ . The facilitated emissions costs  $C_{\text{fac,emissions}}^P$  and  $C_{\text{fac,emissions}}^{\text{no}}$  (§B.4.2) are intermediate quantities; their difference enters the benefit total as  $B_{\text{avoided,emissions}}^P$  (§C.3).

## C Benefits

### C.1 Remedial Benefits ( $B_{\text{remedial}}$ )

Remedial benefits are relief of pre-existing system harms via congestion reduction.  $B_{\text{remedial}}^P = B_{\text{congestion,lifetime}}^P$ .

### C.1.1 Congestion reduction benefit

The annual congestion benefit  $B_{\text{congestion,annual}}$  is defined in §A.4; this subsection adds lifetime aggregation.

| Variable                                 | Description                                                                                          |
|------------------------------------------|------------------------------------------------------------------------------------------------------|
| $B_{\text{congestion,annual}}$           | Annual benefit from congestion reduction (\$/year). From §A.4.                                       |
| $T_{\text{lifetime}}$                    | Project lifetime (years).                                                                            |
| $T_{\text{COD}}$                         | Commercial operation date (year index) (years). First year of operation; benefit stream starts here. |
| $r_{\text{WACC,real}}$                   | Real WACC (decimal). Used to discount benefits (§A.2).                                               |
| $g_{\text{benefit}}$                     | Benefit price escalation rate (decimal). From §A.2.                                                  |
| $B_{\text{congestion,lifetime,nominal}}$ | Nominal lifetime congestion benefit (\$).                                                            |
| $B_{\text{congestion,lifetime}}$         | Real (PV) lifetime congestion benefit (\$).                                                          |

#### Variables

**Equations** As in §A.4,  $B_{\text{congestion,annual}} = H \times \min(\Delta C_{\text{effective}}, X) \times \gamma_{\text{congestion}}$ .

**1. Nominal lifetime congestion benefit.** Growing annual benefit at rate  $g_{\text{benefit}}$  over  $T_{\text{lifetime}}$  years:

$$B_{\text{congestion,lifetime,nominal}} = B_{\text{congestion,annual}} \times \frac{(1 + g_{\text{benefit}})^{T_{\text{lifetime}}} - 1}{g_{\text{benefit}}}.$$

At  $g_{\text{benefit}} = 0$  this reduces to  $B_{\text{congestion,annual}} \times T_{\text{lifetime}}$ .

**2. Real (present value) lifetime congestion benefit.** Growing annual benefit from  $T_{\text{COD}}$  for  $T_{\text{lifetime}}$  years, discounted at  $r_{\text{WACC,real}}$ :

$$B_{\text{congestion,lifetime}} = \frac{B_{\text{congestion,annual}} \times \frac{1 - \left( \frac{1 + g_{\text{benefit}}}{1 + r_{\text{WACC,real}}} \right)^{T_{\text{lifetime}}}}{r_{\text{WACC,real}} - g_{\text{benefit}}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

At  $g_{\text{benefit}} = 0$  this reduces to the flat-annuity formula.

*Link: Effective relief  $\Delta C_{\text{effective}}$ , constrained hours  $H$ , exceedance  $X$ , and  $\gamma_{\text{congestion}}$  are defined in §A.4.*

## C.2 Enabling Benefits ( $B_{\text{enabling}}$ )

Enabling benefits are new productive value the line creates: the benefit of delivered energy.  $B_{\text{enabling}}^P = B_{\text{delivered,lifetime}}^P$ .

### C.2.1 Benefit of delivered energy

Deliverable capacity equals effective capacity as in §A.4: for greenfield,  $\Delta C_{\text{effective}} = \phi \times C_{\text{new}}$ ; for reconductoring or rebuild,  $\Delta C_{\text{effective}} = C_{\text{new}} - C_{\text{old}}$ . This captures the value of additional energy the project enables, including any headroom beyond clearing the constraint. Delivered energy values all throughput, including MWh that would have been curtailed absent the line. Curtailment is not modeled as a separate benefit because it would double-count MWh already captured here at a similar price. Congestion relief prices a different economic object (redispatch cost savings) and is therefore retained as the sole remedial benefit.

| Variable                                | Description                                                                                                                                |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| $\Delta C_{\text{effective}}$           | Effective capacity relief (MW). From §A.4. Greenfield: $\phi C_{\text{new}}$ ; reconductoring/rebuild: $C_{\text{new}} - C_{\text{old}}$ . |
| $u$                                     | Line utilization (dimensionless). From §B.2.4. $0 \leq u \leq 1$ .                                                                         |
| $H$                                     | Hours per year (h/year). e.g. 8760. From §B.2.4.                                                                                           |
| $v_{\text{load}}$                       | Value of load (\$/MWh). Marginal value of energy to end-use load. From §B.2.5.                                                             |
| $E_{\text{delivered,annual}}$           | Deliverable energy per year (MWh/year).                                                                                                    |
| $B_{\text{delivered,annual}}$           | Annual benefit from delivered energy (\$/year).                                                                                            |
| $T_{\text{lifetime}}$                   | Project lifetime (years).                                                                                                                  |
| $T_{\text{COD}}$                        | Commercial operation date (year index) (years). First year of operation; benefit stream starts here.                                       |
| $r_{\text{WACC,real}}$                  | Real WACC (decimal). Used to discount benefits (§A.2).                                                                                     |
| $g_{\text{benefit}}$                    | Benefit price escalation rate (decimal). From §A.2.                                                                                        |
| $B_{\text{delivered,lifetime,nominal}}$ | Nominal lifetime benefit from delivered energy (\$).                                                                                       |
| $B_{\text{delivered,lifetime}}$         | Real (PV) lifetime benefit from delivered energy (\$).                                                                                     |

#### Variables

## Equations

### 1. Deliverable energy (annual).

$$E_{\text{delivered,annual}} = \Delta C_{\text{effective}} \times u \times H.$$

### 2. Annual benefit.

$$B_{\text{delivered,annual}} = E_{\text{delivered,annual}} \times v_{\text{load}}.$$

### 3. Nominal lifetime benefit. Growing annual benefit at rate $g_{\text{benefit}}$ over $T_{\text{lifetime}}$ years:

$$B_{\text{delivered,lifetime,nominal}} = B_{\text{delivered,annual}} \times \frac{(1 + g_{\text{benefit}})^{T_{\text{lifetime}}} - 1}{g_{\text{benefit}}}.$$

At  $g_{\text{benefit}} = 0$  this reduces to  $B_{\text{delivered,annual}} \times T_{\text{lifetime}}$ .

### 4. Real (present value) lifetime benefit. Growing annual benefit from $T_{\text{COD}}$ for $T_{\text{lifetime}}$ years, discounted at $r_{\text{WACC,real}}$ :

$$B_{\text{delivered,lifetime}} = \frac{B_{\text{delivered,annual}} \times \frac{1 - \left( \frac{1 + g_{\text{benefit}}}{1 + r_{\text{WACC,real}}} \right)^{T_{\text{lifetime}}}}{r_{\text{WACC,real}} - g_{\text{benefit}}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

At  $g_{\text{benefit}} = 0$  this reduces to the flat-annuity formula.

#### Default calibration

$g_{\text{benefit}} = 0.01$  (1% real/yr). This reflects an older CAISO Transmission Economic Assessment Methodology (TEAM) practice of applying 1% real escalation to benefits extrapolated beyond the production-cost simulation horizon (Chang, Pfeifenberger, and Hagerty 2013). CAISO's current (2024–2025 Transmission Plan) methodology uses  $g_{\text{benefit}} = 0$  for post-horizon benefit extension (California Independent System Operator 2025). Historical real U.S. electricity prices were flat to declining 2010–2024 (Lam 2026), but FORGE applies escalation uniformly across the entire 30–50 year project lifetime (no simulated-year base), making a conservative positive default appropriate. Sensitivity: 0–4% (Figure F9).

*Link:  $\Delta C_{\text{effective}}$  is defined in §A.4;  $v_{\text{load}}$  is defined in §B.2.5. Total societal benefits comprise  $B_{\text{remedial}}$  (congestion),  $B_{\text{enabling}}$  (delivered energy + capacity value), and  $B_{\text{avoided,emissions}}$  (displacement); see §D for category definitions.*

## C.2.2 Capacity value benefit

Capacity value is the resource-adequacy contribution of incremental transfer capability: the reduction in required local firm capacity (or capacity payments) enabled by dependable imports during system-stress hours. It is economically distinct from delivered energy, which values annual throughput (\$/MWh); capacity value prices the marginal reliability contribution during scarcity hours (\$/MW-yr). The parametric formulation follows Chang, Pfeifenberger, and Hagerty (2013), who recommend valuing deferred generation capacity at Net CONE (Cost of New Entry net of expected energy and ancillary service margins) to avoid double-counting with production cost savings already captured by the congestion and delivered energy benefit streams. A capacity credit factor  $\kappa$  derates nameplate transfer capability to reflect the line's effective load-carrying capability (ELCC) during peak-risk hours, following the empirical range reported in Brooks, Silverstein, and Gramlich (2025) for 73 interregional transmission facilities. An applicability gate  $\mathbb{K}_{\text{cap}}$  defaults to zero.

## Variables

| Variable                       | Description                                                                                                                                                                                                    |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Delta C_{\text{effective}}$  | Effective capacity relief (MW). From §A.4.                                                                                                                                                                     |
| $\kappa$                       | Capacity credit factor (dimensionless, [0, 1]). Fraction of $\Delta C_{\text{effective}}$ accredited for resource adequacy. From published ELCC studies; default 0.                                            |
| $p_{\text{cap}}$               | Capacity price (\$/MW-yr). Net CONE or multi-year average capacity auction clearing price. Must be net of expected energy and ancillary service margins to avoid double-counting with $B_{\text{delivered}}$ . |
| $\mathbb{K}_{\text{cap}}$      | Applicability gate (binary, 0 or 1). Set to 1 only when the receiving region is capacity-short and existing import capability is fully utilized for resource adequacy. Default 0.                              |
| $g_{\text{benefit}}$           | Real annual benefit price escalation rate (decimal). Same as delivered energy.                                                                                                                                 |
| $r_{\text{WACC,real}}$         | Real WACC (decimal).                                                                                                                                                                                           |
| $T_{\text{delay}}$             | Delay years.                                                                                                                                                                                                   |
| $T_{\text{construction}}$      | Construction years.                                                                                                                                                                                            |
| $T_{\text{lifetime}}$          | Project lifetime (years).                                                                                                                                                                                      |
| $B_{\text{capacity,annual}}$   | Annual capacity value benefit (\$/yr).                                                                                                                                                                         |
| $B_{\text{capacity,lifetime}}$ | Lifetime capacity value benefit, real PV (\$). Enters benefit category $B_{\text{enabling}}$ .                                                                                                                 |

## Equations

### 1. Annual capacity value benefit.

$$B_{\text{capacity,annual}} = \Delta C_{\text{effective}} \times \kappa \times p_{\text{cap}} \times \mathbb{K}_{\text{cap}}.$$

### 2. Nominal lifetime benefit.

$$B_{\text{capacity,nominal}} = \sum_{t=1}^{T_{\text{lifetime}}} B_{\text{capacity,annual}} \times (1 + g_{\text{benefit}})^{t-1}.$$

At  $g_{\text{benefit}} = 0$  this reduces to  $B_{\text{capacity,annual}} \times T_{\text{lifetime}}$ .

### 3. Real (present value) lifetime benefit.

Growing annual benefit from  $T_{\text{COD}}$  for  $T_{\text{lifetime}}$  years, discounted at  $r_{\text{WACC,real}}$ :

$$B_{\text{capacity,lifetime}} = \frac{B_{\text{capacity,annual}} \times \frac{1 - \left( \frac{1 + g_{\text{benefit}}}{1 + r_{\text{WACC,real}}} \right)^{T_{\text{lifetime}}}}{r_{\text{WACC,real}} - g_{\text{benefit}}}}{(1 + r_{\text{WACC,real}})^{T_{\text{delay}} + T_{\text{construction}}}}.$$

At  $g_{\text{benefit}} = 0$  this reduces to the flat-annuity formula.

#### Default calibration

$\kappa = 0$  (no capacity credit by default). Override with a published ELCC value from a comparable facility: Brooks, Silverstein, and Gramlich (2025) report accredited capacity ranging from 13% (MISO accreditation of North Plains Connector) to 96% (Soo Green HVDC) of nameplate transfer capability, with dispatchable HVDC interties into capacity-constrained markets typically in the 60–93% range.  $p_{\text{cap}}$ : use the receiving region's published capacity market clearing price or Net CONE. PJM 2026/27 Net CONE reference: ~\$60,400/MW-yr (PJM Interconnection 2025); MISO 2026/27 PRA: ~\$46,000/MW-yr (North/Central, annualized) (MISO 2026). For regions without organized capacity markets (e.g. ERCOT), set  $\mathbb{K}_{\text{cap}} = 0$ .  $\mathbb{K}_{\text{cap}} = 0$  (off by default).

*Link:  $\Delta C_{\text{effective}}$  is defined in §A.4.  $B_{\text{capacity,lifetime}}$  enters the enabling benefit category  $B_{\text{enabling}}$  alongside  $B_{\text{delivered,lifetime}}$ . Net CONE pricing avoids double-counting with the energy margin already captured by  $B_{\text{delivered}}$  (Chang, Pfeifenberger, and Hagerty 2013). This is a parametric screening proxy, not an LOLE/RPM simulation; see Limitation 3.*

## C.3 Displacement avoided emissions

The avoided-emissions benefit  $B_{\text{avoided,emissions}}^P$  is the difference between the no-line and with-line absolute facilitated emissions costs from §B.4.2. Under the single-trajectory grid model (§B.4.2), both trajectories start from the identical grid state at COD and diverge only afterward, as the line's influence ( $g_j^{\text{post}}$ ) pulls the grid onto a different path than it would have followed on its own ( $g_j^{\text{pre}}$ ). Transmission projects that facilitate greater enablement of clean generation leading to displacement of unabated fossil fuels produce a measurable reduction in the social cost of emissions;  $B_{\text{avoided,emissions}}^P$  quantifies that displacement in social-cost terms. A positive  $B_{\text{avoided,emissions}}^P$  means the with-line grid trajectory is cleaner than the without-line counterfactual—an avoided externality cost that enters the benefit category  $B_{\text{avoided,emissions}}$ .

## Variables

| Variable                               | Description                                                                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $C_{\text{fac,emissions}}^P$           | Facilitated emissions cost, project (\$). From §B.4.2.                                                                                                           |
| $C_{\text{fac,emissions}}^{\text{no}}$ | Facilitated emissions cost, no-line (\$). From §B.4.2.                                                                                                           |
| $E_{k,\tau}^{(s)}$                     | Emissions mass under instance $s$ , year $\tau$ (kg). From §B.4.2.                                                                                               |
| $E_{k,\tau}^{\text{avoid}}$            | Avoided emissions mass in year $\tau$ (kg). $= E_{k,\tau}^{\text{no}} - E_{k,\tau}^{\text{with}}$ .                                                              |
| $c_k$                                  | Societal cost per kg of pollutant $k$ (\$/kg). Same as §B.4.1.                                                                                                   |
| $C_{k,\tau}^{\text{avoid}}$            | Avoided emissions cost in year $\tau$ (\$).                                                                                                                      |
| $r_{\text{social}}$                    | Social discount rate (decimal). Same as §B.4.1.                                                                                                                  |
| $T_{\text{COD}}$                       | Commercial operation date (year).                                                                                                                                |
| $T_{\text{lifetime}}$                  | Project lifetime (years).                                                                                                                                        |
| $PV_{\text{avoid}}^k$                  | PV avoided cost for pollutant $k$ (\$).                                                                                                                          |
| $B_{\text{avoided,emissions}}^P$       | Avoided-emissions benefit (\$). $= C_{\text{fac,emissions}}^{\text{no}} - C_{\text{fac,emissions}}^P$ . Enters benefit category $B_{\text{avoided,emissions}}$ . |

## Equations

### 1. Avoided emissions mass per year.

For pollutant  $k$ , using the per-instance emissions mass from §B.4.2:

$$E_{k,\tau}^{\text{avoid}} = E_{k,\tau}^{\text{no}} - E_{k,\tau}^{\text{with}} = E_{\text{delivered,annual}} \times (EF_k^{\text{no}}(\tau) - EF_k^{\text{with}}(\tau)).$$

Positive  $E_{k,\tau}^{\text{avoid}}$  means fewer emissions under the with-line grid trajectory than under the without-line counterfactual. Negative values (with-line trajectory dirtier) are allowed.

## 2. Avoided emissions cost per year.

$$C_{k,\tau}^{\text{avoid}} = E_{k,\tau}^{\text{avoid}} \times c_k(\tau).$$

## 3. PV of avoided cost per pollutant.

$$PV_k^{\text{avoid}} = PV_k^{\text{no}} - PV_k^{\text{with}} = \sum_{\tau=1}^{T_{\text{lifetime}}} \frac{C_{k,\tau}^{\text{avoid}}}{(1 + r_{\text{social}})^{T_{\text{COD}} + \tau - 1}}.$$

## 4. Total avoided-emissions benefit.

$$B_{\text{avoided,emissions}}^P = C_{\text{fac,emissions}}^{\text{no}} - C_{\text{fac,emissions}}^P = \sum_k PV_k^{\text{avoid}}.$$

Positive = cleaner with-line trajectory = avoided externality.  $B_{\text{avoided,emissions}}^P$  enters  $B_{\text{avoided,emissions}}$  (§D).

**Pairwise corollary.**  $\Delta B_{\text{avoided,emissions}}^{P:B} = B_{\text{avoided,emissions}}^P - B_{\text{avoided,emissions}}^B = C_{\text{fac,emissions}}^B - C_{\text{fac,emissions}}^P$  (the no-line reference term cancels). See §D for incremental reporting.

### C.3.1 Displacement Delay Emissions Cost

Each year of project delay represents a year of foregone emissions displacement. Under the single-trajectory grid model (§B.4.2), both trajectories start from the same initial grid shares  $p_j(0)$  at year 0: the without-line path evolves at  $g_j^{\text{pre}}$ , continuing the trajectory that is later used as the grid's actual state at COD; the with-line path evolves *hypothetically* at  $g_j^{\text{post}}$ , as if the line's influence on the grid had begun at year 0 — "what if the line had been built on time?" The displacement delay cost values the foregone benefit of that hypothetical, computing for each delay year the emissions reduction that would have occurred had the project been operational since year 0:

$$C_{\text{displacement,delay}} = \sum_{t=1}^{T_{\text{delay}}} \frac{B_{\text{displacement}}(t)}{(1 + r_{\text{social}})^t},$$

where

$$B_{\text{displacement}}(t) = E_{\text{delivered}} \times \sum_k \left[ \sum_j (p_j^{\text{no}}(t) - p_j^{\text{with}}(t)) \times I_{j,k} \right] \times c_k(t),$$

$$p_j^{\text{no}}(t) \propto p_j(0)(1 + g_j^{\text{pre}})^{t-1}, \quad p_j^{\text{with}}(t) \propto p_j(0)(1 + g_j^{\text{post}})^{t-1} \quad (\text{both renormalized each year}).$$

where  $c_k(0)$  is the base-year social cost (Rennert et al. 2022; Office of Management and Budget 2023) and  $g_k$  is the annual real growth rate calibrated to damage-pathway projections, so  $c_k(t) = c_k(0) \times (1 + g_k)^t$ .

**Single-trajectory handoff at COD.** At COD, the hypothetical with-line path used above is discarded and replaced by the grid's actual state,  $p_j(T_{\text{COD}}) = p_j^{\text{no}}(T_{\text{COD}})$  — the grid *is* whatever it evolved into without the line during delay plus construction, since the line has not yet influenced it (§B.4.2, Eq. 2). The operational-period displacement benefit (§C.3) then evolves both the with-line and without-line paths forward from that identical COD state, rather than restarting from year-0 percentages. Consequently, a longer delay leaves the grid further along its without-line decarbonization path by the time the project reaches COD, which mechanically reduces the operational displacement benefit available to a delayed project relative to one that reached COD sooner — consistent with the expectation that a cleaner starting grid has less fossil generation to displace.

## D Reporting Framework

This section assembles the component present values defined in the four cost category subsections (§B.1–§B.4) into two reporting scalars (scenario-internal net benefit and incremental net benefit vs a stated baseline) and defines the sign conventions used throughout the paper.

## Category totals and grand cost

Cost components are organized into four reporting categories (defined in §B.1.3, §B.2.7, §B.3.2, and §B.4):

| Category               | Equation                                                         | Constituent components                      |
|------------------------|------------------------------------------------------------------|---------------------------------------------|
| $C_{\text{hard}}$      | $C_{\text{build}} + C_{\text{ROW,capital}} + C_{\text{env.mit}}$ | §B.1.1, §B.1.2, §B.1.3                      |
| $C_{\text{soft}}$      | $C_{\text{operational}} + C_{\text{energy}} + C_{\text{delay}}$  | §B.2.1–§B.2.3, §B.2.4–§B.2.5, §B.2.6–§C.3.1 |
| $C_{\text{risk}}$      | $C_{\text{wf}} + C_{\text{outage}}$                              | §B.3                                        |
| $C_{\text{emissions}}$ | $C_{\text{comp,emissions}}$                                      | §B.4.1                                      |

Grand total cost for scenario  $P$ :

$$C^P = C_{\text{hard}}^P + C_{\text{soft}}^P + C_{\text{risk}}^P + C_{\text{emissions}}^P.$$

$C_{\text{emissions}}$  ( $= C_{\text{comp,emissions}}$ ) includes loss-compensation emissions only (§B.4.1). The facilitated emissions costs  $C_{\text{fac,emissions}}^P$  and  $C_{\text{fac,emissions}}^{\text{no}}$  are intermediate quantities (§B.4.2); their difference enters the benefit total as  $B_{\text{avoided,emissions}}^P$  (§C.3).

## Benefit categories and total benefit

The benefit streams from §C are organized into three reporting categories:

| Category                       | Equation                                                            | Constituent components |
|--------------------------------|---------------------------------------------------------------------|------------------------|
| $B_{\text{remedial}}$          | $B_{\text{congestion,lifetime}}$                                    | §C.1.1                 |
| $B_{\text{enabling}}$          | $B_{\text{delivered,lifetime}} + B_{\text{capacity,lifetime}}$      | §C.2.1, §C.2.2         |
| $B_{\text{avoided,emissions}}$ | $C_{\text{fac,emissions}}^{\text{no}} - C_{\text{fac,emissions}}^P$ | §C.3                   |

Total benefit for scenario  $P$ :

$$B^P = B_{\text{remedial}}^P + B_{\text{enabling}}^P + B_{\text{avoided,emissions}}^P.$$

$B_{\text{remedial}}$  captures relief of pre-existing system harms (binding congestion).  $B_{\text{enabling}}$  captures new productive value (delivered energy throughput, including MWh that would have been curtailed absent the line).  $B_{\text{avoided,emissions}}$  captures the clean-generation incentive: the reduction in societal emissions costs when the project’s facilitated mix displaces a dirtier counterfactual (§C.3).

## Scenario-internal net benefit

The scenario-internal net benefit answers “how does scenario  $P$  look on its own ledger?”:

$$\text{NB}^P = B^P - C^P.$$

## Incremental comparisons

Pairwise comparisons follow an avoided-cost sign convention:  $\Delta C^{P:B} = C^B - C^P$  (positive means scenario  $P$  has lower cost than baseline  $B$ ). Benefits difference normally:  $\Delta B^{P:B} = B^P - B^B$ . Incremental net benefit is  $\Delta \text{NB}^{P:B} = \Delta B^{P:B} + \Delta C^{P:B} = \text{NB}^P - \text{NB}^B$ ; these definitions apply componentwise to each category. A lower risk cost under  $P$  shows as positive  $\Delta C_{\text{risk}}^{P:B}$  (avoided risk), not as additional benefit. The emissions cost category holds loss-compensation emissions only ( $C_{\text{comp,emissions}}^P$ ); displacement enters the benefit total as  $B_{\text{avoided,emissions}}^P$ .

### D.1 Benefit-cost ratios

A benefit-cost ratio under perspective  $\pi$  is:

$$\text{BCR}_{\pi} = \frac{B_{\pi}^P}{C_{\pi}^P}.$$

$\text{BCR}_{\pi} > 1$  means benefits exceed costs under that perspective. FORGE reports three core perspectives and six exclusion variants. Each answers a different question for a different audience; the choice of which BCR to prioritize depends on the stakeholder’s position relative to the project.

Table 5: Core BCR perspectives. Each answers a different question for a different audience; see text below for definitions.

| Perspective     | Label                      | Numerator ( $B_\pi$ )                           | Denominator ( $C_\pi$ )                           |
|-----------------|----------------------------|-------------------------------------------------|---------------------------------------------------|
| <i>Societal</i> |                            |                                                 |                                                   |
| Societal        | $\text{BCR}_{\text{soc}}$  | $B^P$                                           | $C^P$                                             |
| <i>Firm</i>     |                            |                                                 |                                                   |
| Utility / TSP   | $\text{BCR}_{\text{util}}$ | $\text{ATRR}_{\text{PV}}$                       | $\text{ATRR}_{\text{PV}} + C_{\text{base delay}}$ |
| Ratepayer       | $\text{BCR}_{\text{rp}}$   | $B_{\text{remedial}}^P + B_{\text{enabling}}^P$ | $\text{ATRR}_{\text{PV}} + C_{\text{loss}}$       |

Table 6: Societal BCR exclusion variants. Each removes specific cost or benefit categories from the societal benchmark.

| Perspective             | Label                              | Numerator ( $B_\pi$ )                  | Denominator ( $C_\pi$ )                   |
|-------------------------|------------------------------------|----------------------------------------|-------------------------------------------|
| Excl. avoided emissions | $\text{BCR}_{\text{excl av.em.}}$  | $B^P - B_{\text{avoided,emissions}}^P$ | $C^P$                                     |
| Excl. emissions costs   | $\text{BCR}_{\text{excl em.cost}}$ | $B^P$                                  | $C^P - C_{\text{emissions}}$              |
| Excl. all emissions     | $\text{BCR}_{\text{excl all em.}}$ | $B^P - B_{\text{avoided,emissions}}^P$ | $C^P - C_{\text{emissions}}$              |
| Excl. outage            | $\text{BCR}_{\text{excl out}}$     | $B^P$                                  | $C^P - C_{\text{outage}}$                 |
| Excl. wildfire          | $\text{BCR}_{\text{excl wf}}$      | $B^P$                                  | $C^P - C_{\text{wf}}$                     |
| Excl. wildfire + outage | $\text{BCR}_{\text{excl wf+out}}$  | $B^P$                                  | $C^P - C_{\text{wf}} - C_{\text{outage}}$ |

The following paragraphs justify each perspective, cite the regulatory and academic authority for its inclusion, and explain when  $\text{BCR} < 1$  does and does not signal project failure.

**Societal ( $\text{BCR}_{\text{soc}}$ ).** The societal BCR aggregates all quantifiable benefits against all costs. This is the broadest reading of FERC Order No. 1920’s requirement that benefits be “evaluated comprehensively over a minimum 20-year horizon” (FERC 2024). Pfeifenberger recommends that planners “account for the full range of transmission projects’ benefits and use multi-value planning to comprehensively identify investments that cost-effectively address all categories of needs and benefits” (Pfeifenberger 2022). Revenue is excluded from the societal BCR because it is a transfer payment between utility and ratepayers—zero-sum from society’s perspective (cf. (Pfeifenberger et al. 2021)).  $\text{BCR}_{\text{soc}} < 1$  means that, absent unquantified benefits (reliability, resource adequacy, option value), societal costs outweigh societal benefits.

**Numerator discipline.** Societal BCRs use  $B^P$  in the numerator. Gains from lower risk, emissions, capital, or delay show up as lower  $C_\pi$  in the denominator—not as extra benefit lines in the numerator. This is the same principle as the incremental avoided-cost convention: a positive  $\Delta C^{P:B}$  means  $P$  avoids cost, not that  $P$  earns additional benefit. The full revenue requirement ( $\text{ATRR}_{\text{PV}}$ , §E) is a transfer; it enters perspective-specific views but not the societal NB.

**Utility / TSP ( $\text{BCR}_{\text{util}}$ ).** The Utility BCR is a financial viability test from the utility’s perspective: does the revenue the utility collects cover everything the utility pays? Both numerator and denominator use the utility’s own cost basis—no societal PV. The numerator is the full annual transmission revenue requirement ( $\text{ATRR}_{\text{PV}}$ : capital recovery + O&M + insurance + ROW rent), following the FERC cost-of-service formula (FERC 2026). The denominator is  $\text{ATRR}_{\text{PV}} + C_{\text{base delay}}$ : the same recoverable costs, plus pre-construction delay expenditures (legal, admin, permitting) that are not in rate base. Since the  $\text{ATRR}$  terms cancel:

$$\text{BCR}_{\text{util}} = \frac{\text{ATRR}_{\text{PV}}}{\text{ATRR}_{\text{PV}} + C_{\text{base delay}}} = 1.0 \text{ when } C_{\text{base delay}} = 0.$$

$\text{BCR}_{\text{util}} = 1.0$  is a structural identity for zero-delay regulated projects, not an approximation. Deviation below 1.0 measures exactly the fraction of the utility’s costs that the revenue requirement does not recover—the pre-COD delay penalty (FERC 1995).

**Utility denominator scope.** The utility denominator includes only costs the utility bears: the revenue requirement ( $\text{ATRR}$ , which covers capital recovery, O&M, insurance, and ROW rent) plus base delay costs ( $C_{\text{base delay}}$ : legal, admin, labor, regulatory, project management during the pre-construction period). Congestion delay costs—the societal opportunity cost of constraints persisting during the delay period—are excluded because the utility does not bear them; they flow to market participants through locational marginal prices (LMP).

**Ratepayer ( $\text{BCR}_{\text{rp}}$ ).** The Ratepayer BCR measures whether the benefits that flow to customer bills—reduced energy procurement costs from congestion relief and delivered energy value—exceed what ratepayers pay through transmission charges and energy losses. The CAISO Transmission Economic Assessment Methodology defines

ratepayer benefit as the “difference of net load payment between the cases pre and post project” (CAISO 2017). Unlike the societal BCRs, the ratepayer numerator excludes avoided emissions ( $B_{\text{avoided,emissions}}^P$ ) because they are a societal externality reduction that does not reduce wholesale energy purchase costs or appear on customer bills absent an embedded carbon price.  $\text{BCR}_{\text{rp}} < 1$  means ratepayers pay more (through rates and losses) than they receive in bill-relevant benefits; this is expected for projects whose primary value is emissions reduction or reliability.

**Exclusion variants.** Six exclusion variants of the societal BCR are reported. Each answers a specific sensitivity question:

1. **Excl. avoided emissions:** FERC Order No. 1920 does not require emissions quantification as a transmission benefit; the seven required benefit categories reflect “objective, measurable changes in transmission system conditions, rather than achievement of state public policies” (FERC 2024). This variant shows the BCR under the FERC-minimum benefit set. It is the relevant metric for jurisdictions that do not recognize a social cost of carbon.
2. **Excl. emissions costs:** Removes loss-compensation emissions from the cost denominator while retaining displacement benefit in the numerator. This isolates the effect of the loss-compensation emissions model for projects where line losses drive a large emissions cost (e.g., long AC lines on fossil-heavy grids). Answers: would this project pass a societal screen if we did not penalize it for its line losses?
3. **Excl. all emissions:** Removes displacement benefit from the numerator AND loss-compensation emissions from the denominator—both sides of the emissions accounting. This is the pure market-flow BCR: only congestion relief and delivered energy in the numerator; only hard, soft, and risk costs in the denominator. Useful for jurisdictions that reject both the social cost of carbon and avoided-emissions crediting.
4. **Excl. outage:** Outage risk is dominated by line length and construction type (single-circuit HVDC vs. meshed AC). This variant isolates the outage cost assumption for projects where the outage model’s calibration is contested while wildfire risk is accepted.
5. **Excl. wildfire:** Wildfire risk is geographically concentrated. Projects in low-wildfire regions (e.g., the Great Plains, Midwest) should not bear wildfire costs in their BCR.
6. **Excl. wildfire + outage:** Removes all probabilistic risk costs, showing the deterministic BCR for stakeholders who question probabilistic risk valuation methodology.

## E Revenue

This section defines capital recovery—the depreciation and return-on-rate-base component of the full FERC revenue requirement. Capital recovery is a transfer such that the utility recovers its capital investment from ratepayers (it is not a cost or benefit to society). The full revenue requirement that enters the utility BCR also includes O&M and insurance (see the preceding paragraph on the full revenue requirement). Both perspectives use the same capital spending (same real-dollar amounts and timing). Utility (regulatory): base-year spending in CWIP is capitalized with AFUDC at real WACC to COD, giving  $RB_{\text{real}}$ . Society: same spending stream is discounted to base year at real WACC; AFUDC does not appear. Same inputs and time value; different purpose (rate base vs. PV of spending).

## Variables

| Variable                | Description                                                                                                                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $RB_{\text{real}}$      | Rate base at COD, base-year (\$). AFUDC capital at COD (see §A.3). Already in base-year dollars.                                                                                                                 |
| $C^{\text{cap}}$        | Total capital in rate base at COD (\$). $RB_{\text{real}} = C^{\text{cap}}$ .                                                                                                                                    |
| $n$                     | Project lifetime (years). Depreciation and revenue-requirement horizon; same as $T_{\text{lifetime}}$ .                                                                                                          |
| $r_{\text{WACC,real}}$  | Real WACC (decimal). Derived from nominal WACC and inflation via the Fisher equation: $r_{\text{WACC,real}} = (1 + r_{\text{WACC,nominal}})/(1 + \pi) - 1$ . Used for both the return component and discounting. |
| $R_t$                   | Annual revenue requirement in year $t$ , real terms (\$/year). Straight-line depreciation plus return on the declining undepreciated balance; see equations below.                                               |
| $T_{\text{lifetime}}$   | Project lifetime (years).                                                                                                                                                                                        |
| $R_{\text{total,real}}$ | Total revenue over lifetime, undiscounted, real (\$). Sum of $R_t$ over $t = 1, \dots, n$ ; no time value.                                                                                                       |
| $R_{\text{PV}}$         | Present value of capital recovery (\$). Capital recovery component of the utility’s revenue requirement; combined with O&M, insurance, and ROW rent PV to form $\text{ATRR}_{\text{PV}}$ .                       |
| $T_{\text{COD}}$        | Commercial operation date (year index). Years from base year to first year of operation.                                                                                                                         |

## Equations

1. **Rate base.**  $RB_{\text{real}}$  is the AFUDC-capitalized rate base from §A.3, already in base-year dollars.
2. **COD (year index from base year).**

$$T_{\text{COD}} = T_{\text{delay}} + T_{\text{construction}} + 1.$$

3. **Annual revenue requirement.** Revenue follows the standard regulatory formulation: straight-line depreciation plus return on the undepreciated rate base at the real weighted-average cost of capital (FERC 2026; FERC 1995):

$$R_t = \frac{RB_{\text{real}}}{n} + RB_{\text{real}} \times r_{\text{WACC,real}} \times \left(1 - \frac{t-1}{n}\right), \quad t = 1, \dots, n.$$

The first term is straight-line depreciation (constant); the second is return on the declining undepreciated balance. Revenue declines from  $R_1 = RB_{\text{real}}(1/n + r_{\text{WACC,real}})$  in year 1 to  $R_n = RB_{\text{real}}/n$  in year  $n$  (FERC 2026).

4. **Total revenue over lifetime (real, undiscounted).**

$$R_{\text{total,real}} = \sum_{t=1}^n R_t.$$

5. **Present value of revenue (benefit to utility / cost to ratepayers).**

$$R_{\text{PV}} = \sum_{t=1}^n \frac{R_t}{(1 + r_{\text{WACC,real}})^{T_{\text{COD}} + t - 1}}.$$

By construction,  $R_{\text{PV}} = RB_{\text{real}}/(1 + r_{\text{WACC,real}})^{T_{\text{COD}} - 1}$  — the NPV neutrality identity ensures the utility exactly recovers its investment plus a fair return (FERC 1995). So  $B_{\text{utility}} = R_{\text{PV}}$  and  $C_{\text{ratepayers}} = B_{\text{utility}} = R_{\text{PV}}$ .

**Full revenue requirement (utility perspective).** The regulated utility’s annual revenue requirement follows FERC cost-of-service principles (FERC 2026):

$$\text{ATRR} = \underbrace{RB \times r + \text{Depreciation}}_{\text{capital recovery}} + \underbrace{C_{\text{O\&M}} + C_{\text{insurance}} + C_{\text{rent,total}}}_{\text{operating expenses}}.$$

O&M expenses (FERC Accounts 560–573) and property insurance (Account 924, allocated via Kansas-Nebraska gross-plant ratios) are recoverable operating expenses passed through to ratepayers at cost (*Cost-of-Service Rates Manual*, pp. 6, 19, 24). The utility BCR numerator includes all three components—capital recovery, O&M, and insurance—reflecting that a regulated utility recovers all prudently incurred costs through its formula rate. See the utility BCR definition above (§D.1) for the zero-delay identity and delay-deviation interpretation.

## F Case-Study Data Sources

The four tables below document the primary data sources used for each case study in ?? and what each source provided to the FORGE framework.

Table 7: AEP LRGV: data sources and what each provided to FORGE.

| Source                                                       | What It Provided                                                                                                                             |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| CTC Global case study / EPRI ACCC micro-site                 | Ampacity increase (77%), conductor specifications, operating experience (CTC Global 2016; EPRI 2026)                                         |
| AEP World Bank presentation (2025) / EEI Edison Award (2016) | Total cost \$375M vs. \$418M ACSR estimate; timeline; scope confirmation (Berger 2025; Edison Electric Institute 2016)                       |
| TD World (2014, 2016) / Quanta Energized Services            | Energized reconductoring and construction methodology (Henningsen and Bryant 2014; Trueblood and Irwin 2016; Quanta Energized Services 2026) |
| AEP Texas PUCT rate case (2024)                              | WACC 6.66%, ROE 9.76%, capital structure 57.5/42.5 D/E (PUCT 2024)                                                                           |
| EIA State Electricity Profiles (2010), Texas                 | Initial grid mix and identical pre-/post-COD evolution rates (U.S. Energy Information Administration 2012)                                   |

Table 8: CTT CREZ: data sources and what each provided to FORGE.

| Source                                     | What It Provided                                                                                                                                           |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CTT PUCT Rate Case, Docket 57467 (2025)    | WACC 6.26%, ROE 9.60%, capital structure 59/41 D/E (PUCT 2025)                                                                                             |
| LS Power CREZ data (NY DPS Case 18-T-0207) | All-in cost \$450M; cost/mile comparison across CREZ builders (LS Power 2018)                                                                              |
| Potomac Economics ERCOT SOM (2011)         | West–North average price spread (\$7.37/MWh); West-to-North constraint binding at 76% utilization (Potomac Economics 2012)                                 |
| EIA ERCOT fuel mix / Brattle Group         | Initial grid mix and curtailment trajectory ( 17% $\rightarrow$ 1.6%) used to parameterize pre-/post-COD grid evolution (EIA 2024; The Brattle Group 2014) |
| EPA eGRID (TRE region)                     | Emission intensities for the Texas grid (EPA 2025a)                                                                                                        |

Table 9: SunZia: data sources and what each provided to FORGE.

| Source                                                              | What It Provided                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAISO PTO application (2024) / BLM Record of Decision               | Conductor spec, capacity (3,021 MW), route length (550 mi), land ownership split (SunZia Transmission, LLC 2024; BLM 2015)                                                                                                                                                                                                                                          |
| NM RETA Transmission Study (2022) / Gridworks Connected West (2024) | Pre-SunZia export limit and Western U.S. curtailment reference case (RETA 2022; Gridworks 2024)                                                                                                                                                                                                                                                                     |
| EIA State Electricity Profiles (2008), AZ+NM                        | Initial grid mix and pre-/post-COD wind and coal evolution rates (U.S. Energy Information Administration 2010)                                                                                                                                                                                                                                                      |
| LADWP–Pattern Energy PPA                                            | Capacity factor (46%) and wind PPA price (\$41/MWh) (Utility Dive 2022)                                                                                                                                                                                                                                                                                             |
| Pattern Energy / RETA / CleanBridge; Heatmap News (2024)            | Total cost, financing, and the 17-year permitting narrative (Pattern Energy Group LP 2023; RETA 2024; CleanBridge 2025; Roberts 2024)                                                                                                                                                                                                                               |
| EPA eGRID (AZNM subregion)                                          | Emission intensities (EPA 2025b)                                                                                                                                                                                                                                                                                                                                    |
| Capacity value inputs                                               | $\kappa = 0.35$ (CPUC MTR Wind NM 2026 ELCC (Energy Division Staff, California Public Utilities Commission 2023); E3/Astrapé 2023 ELCC update (Energy and Environmental Economics and Astrapé Consulting 2023)); $p_{\text{cap}} = \$88,090/\text{MW-yr}$ (CAISO CPM soft offer cap (Corporation 2024), FERC ER24-1225 (Federal Energy Regulatory Commission 2024)) |

Table 10: TBC: data sources and what each provided to FORGE.

| Source                                                                 | What It Provided                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cupertino Electric                                                     | Route, capacity, voltage, and converter technology (Cupertino Electric, Inc. 2026)                                                                                                                                                                                                                                                                                                                                                         |
| IJGlobal (2009) / Prysmian (2007)                                      | Financing structure and total cost (\$505M); cable supply contract (IJGlobal 2009; Prysmian Group 2007)                                                                                                                                                                                                                                                                                                                                    |
| East Bay Times (2007, 2010)                                            | Permitting timeline and COD (East Bay Times 2007; East Bay Times 2010)                                                                                                                                                                                                                                                                                                                                                                     |
| TBC 2024 Wildfire Mitigation Plan / ENTSO-E/Europacable (2019)         | Zero wildfire exposure; submarine cable failure rate and repair cost (Trans Bay Cable LLC 2024; ENTSO-E and Europacable 2019)                                                                                                                                                                                                                                                                                                              |
| McCarthy et al. (2009) / CA RPS & SB 1368 / EPA eGRID (CAMX subregion) | Initial California grid mix (2005, incl. imports), identical pre-/post-COD evolution rates, and emission intensities (McCarthy, Yang, and Ogden 2009; California State Legislature 2006b; California State Legislature 2006a; EPA 2025b)                                                                                                                                                                                                   |
| Capacity value inputs                                                  | $\kappa = 1.0$ (TBC’s 400 MW rating (Cupertino Electric, Inc. 2026) enabled retirement of 360 MW of local generation (Potrero units 3–6) (CAISO 2011), full nameplate credited as local RA capacity; TBC 2025 Availability Report: 99.97% net availability (Trans Bay Cable LLC 2025)); $p_{\text{cap}} = \$88,090/\text{MW-yr}$ (CAISO CPM soft offer cap (Corporation 2024), FERC ER24-1225 (Federal Energy Regulatory Commission 2024)) |

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